FLOWER: training-data action trajectory comparison

27 demonstrations / 9 tasks / 8,023 recorded observations

EMA checkpoint: epoch 20 / step 20,820 | CPU FP32 | 13-D proprioception + 500 Hz FT GRU / AdaLN

Mean XYZ error 1.471 mm P95 5.082 mm Mean rotation error 0.050 deg Gripper sign accuracy 99.1%

Task	Frames	XYZ mean (mm)	XYZ P95 (mm)	Rotation mean (deg)	Gripper accuracy
audio_jack	868	1.956	6.891	0.004	99.2%
barlatch	1044	1.172	3.444	0.139	98.9%
bnc	1105	1.184	3.438	0.062	99.0%
hdmi	828	1.075	3.162	0.003	99.8%
knob	929	1.202	4.340	0.097	99.1%
lan	835	1.454	5.445	0.003	99.0%
outlet	827	1.598	5.243	0.004	99.0%
rotary_switch	779	1.249	4.286	0.107	98.7%
usb	808	2.547	8.395	0.004	99.3%

Selection: task demonstrations 1, 25, 50; collection positions 1, 3, 5; no selection by prediction quality.

The first of 10 predicted actions is decoded at every complete training-window start (4 Euler steps, fixed per-frame noise seed).

Ground truth is the requested target reconstructed from the stored delta action and recorded TCP. Rotation error is geodesic.

Existing 224x224 crops; scaling and CLIP normalization only. No color jitter, second crop, extra state/FT normalization or smoothing.

Original state/FT statistics and symlog-minmax action inverse are used. All velocities remain in the 13-D state; FT is a separate input.

This is recorded-observation replay on training data. Predictions do not change subsequent inputs; these are not robot rollout results.

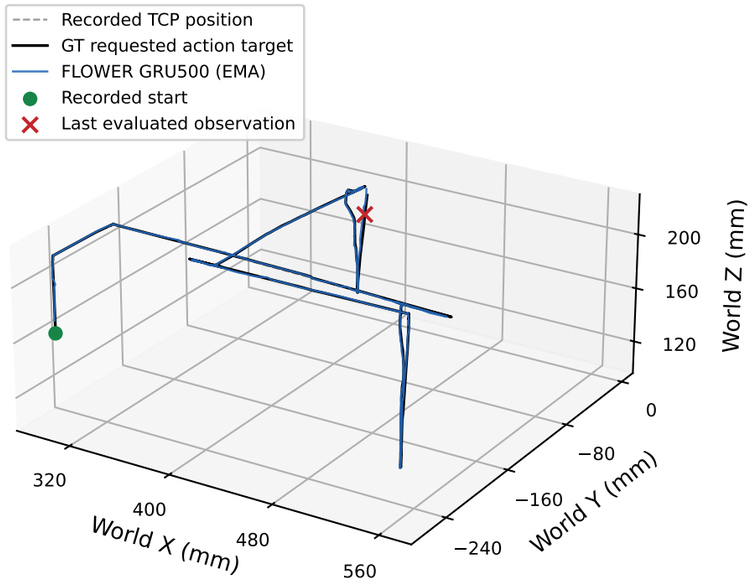
FLOWER GRU500 | audio_jack p1 | episode 1/27 | RAW

1_demos_2026-10-19_17-40-21_330892.pkl

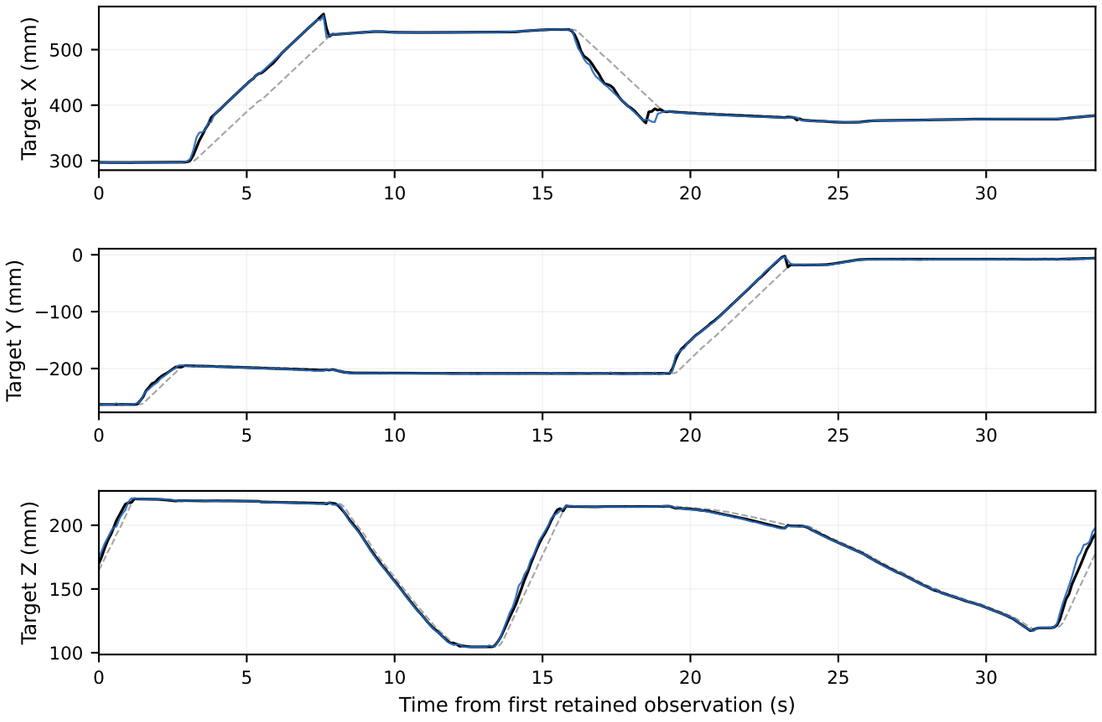
338 evaluated / 347 retained / 349 original frames | source starts at frame 2 | action rate 10 Hz

Mean XYZ target error: 2.092 mm | P95: 8.445 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 98.8%

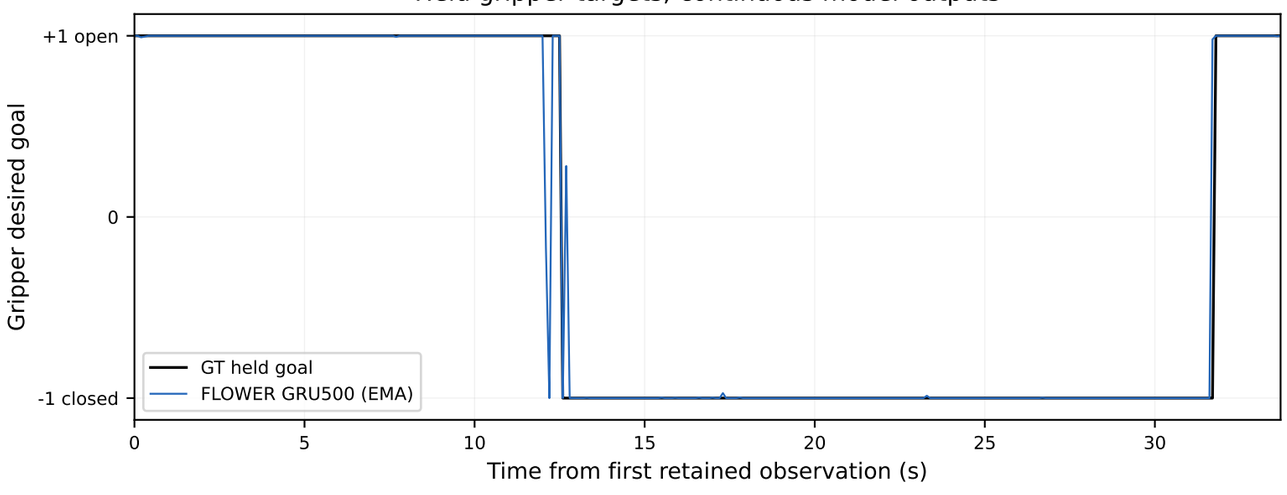
Recorded TCP and predicted command targets



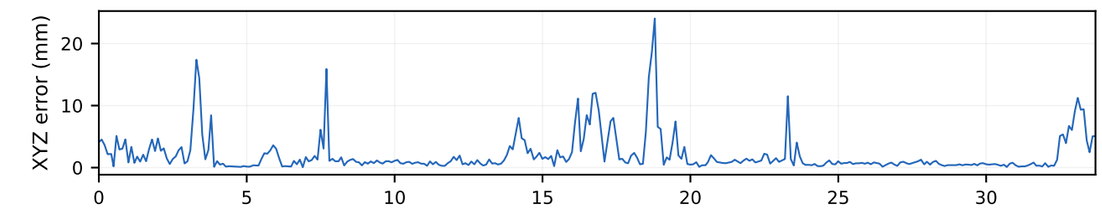
First command from each recorded training observation



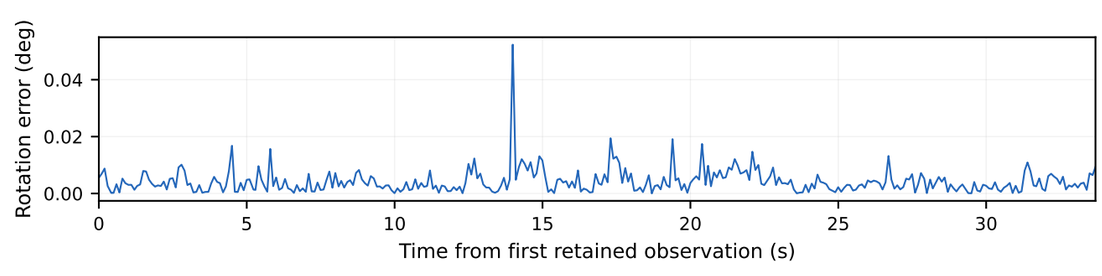
Held gripper targets; continuous model outputs



XYZ error (mm)



Rotation error (deg)



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

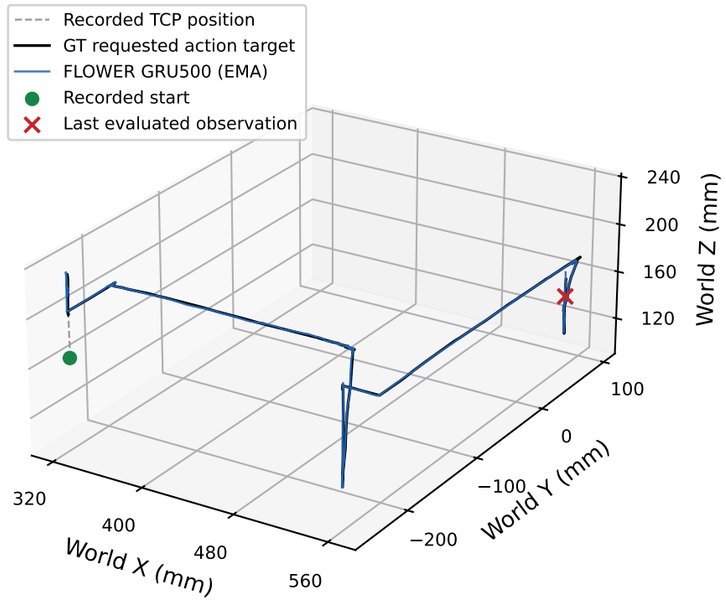
FLOWER GRU500 | audio_jack p3 | episode 2/27 | RAW

1_demos_2026-10-22_16-38-26_059280.pkl

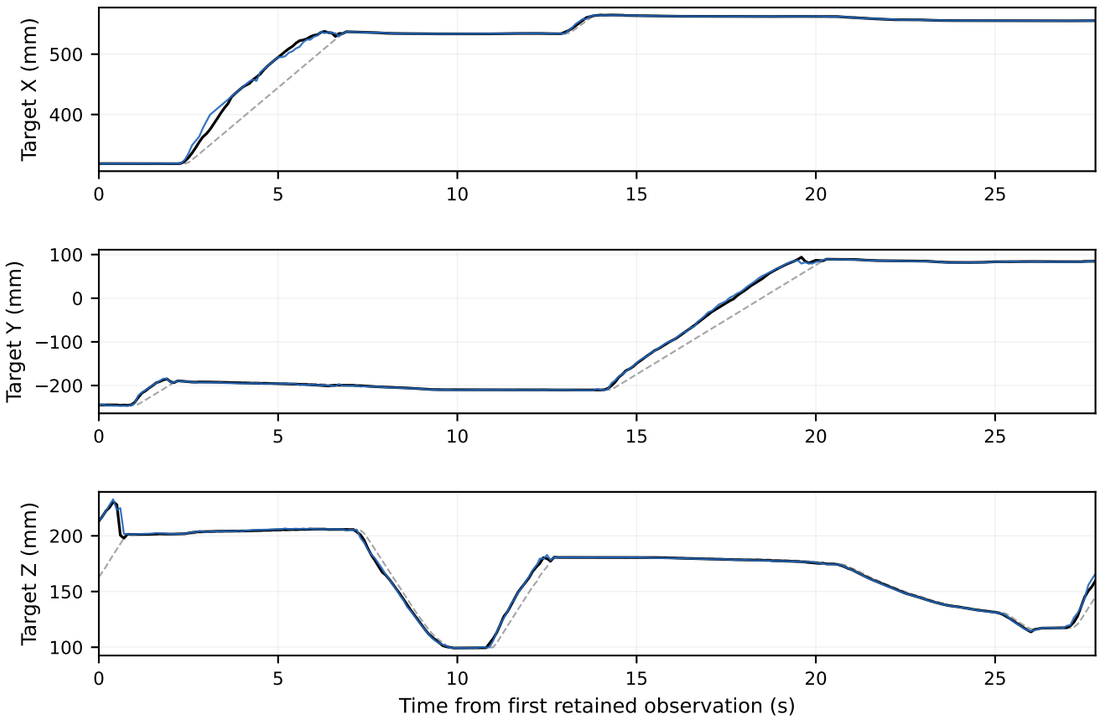
279 evaluated / 288 retained / 289 original frames | source starts at frame 1 | action rate 10 Hz

Mean XYZ target error: 2.366 mm | P95: 8.286 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 99.6%

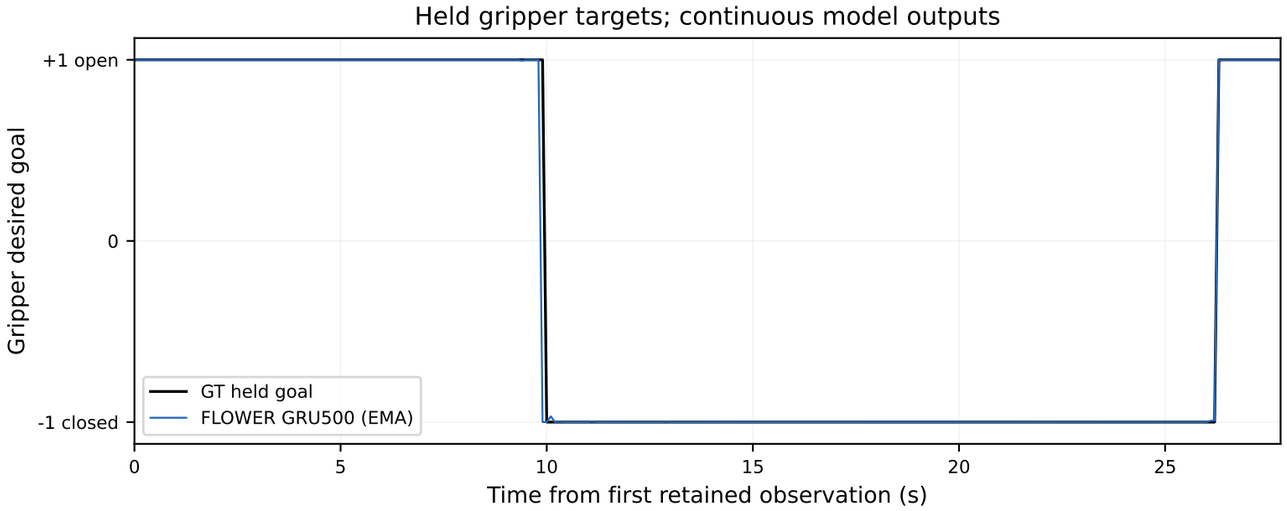
Recorded TCP and predicted command targets



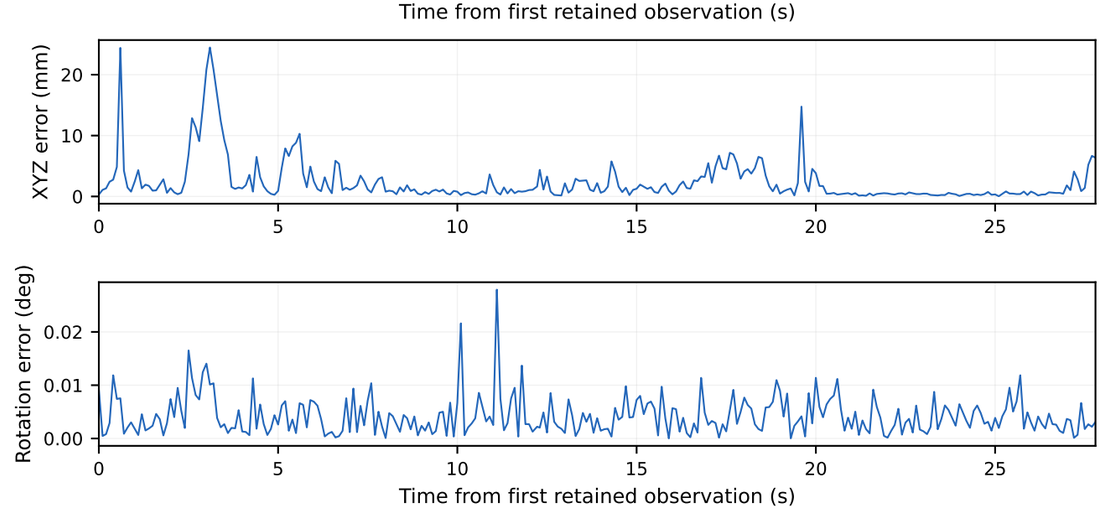
First command from each recorded training observation



Held gripper targets; continuous model outputs



Command error



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

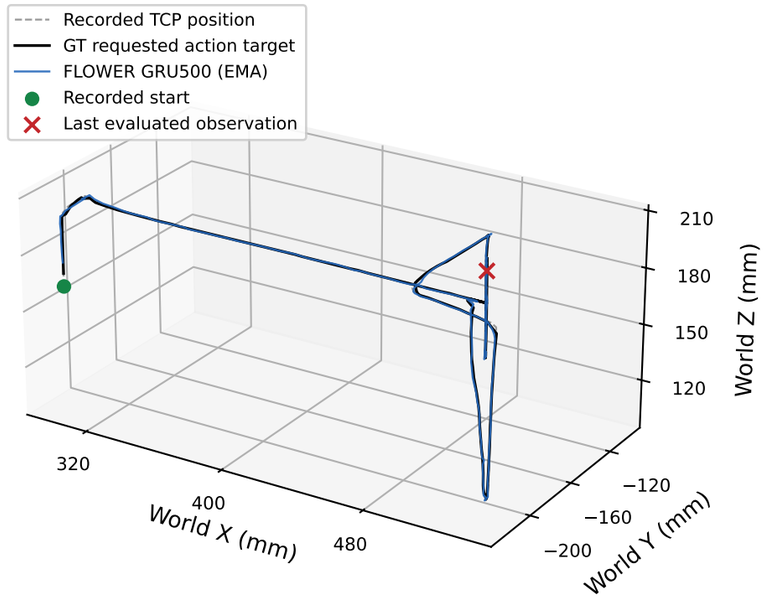
FLOWER GRU500 | audio_jack p5 | episode 3/27 | RAW

1_demos_2026-10-22_20-48-56_643112.pkl

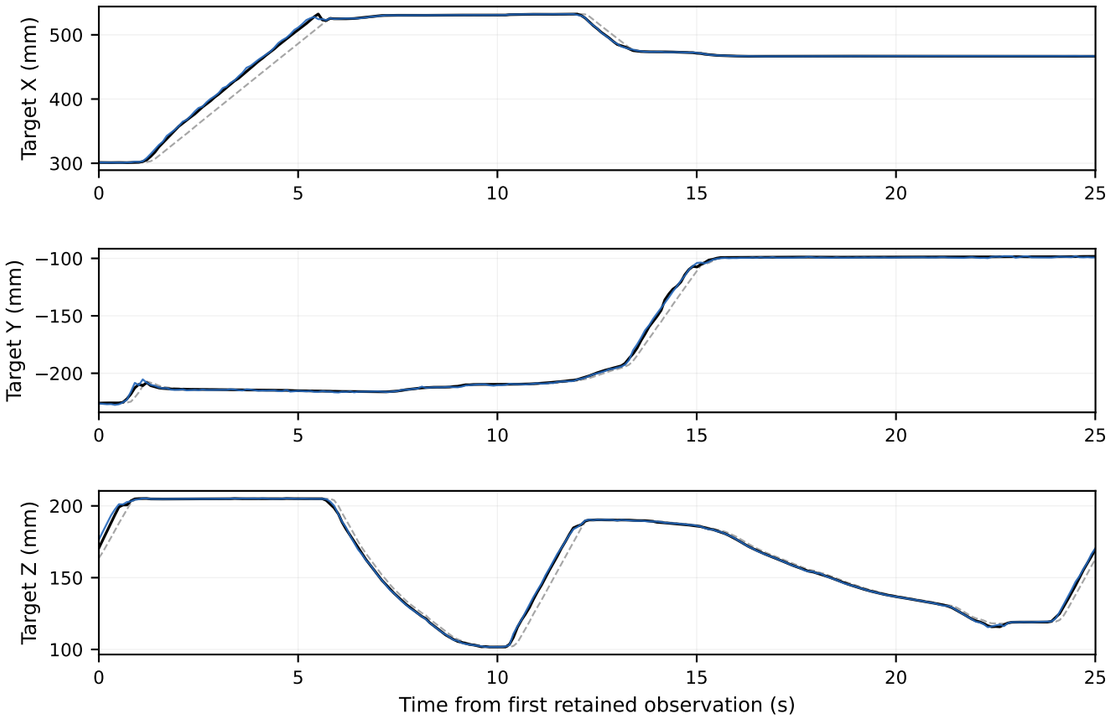
251 evaluated / 260 retained / 266 original frames | source starts at frame 6 | action rate 10 Hz

Mean XYZ target error: 1.319 mm | P95: 4.198 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 99.2%

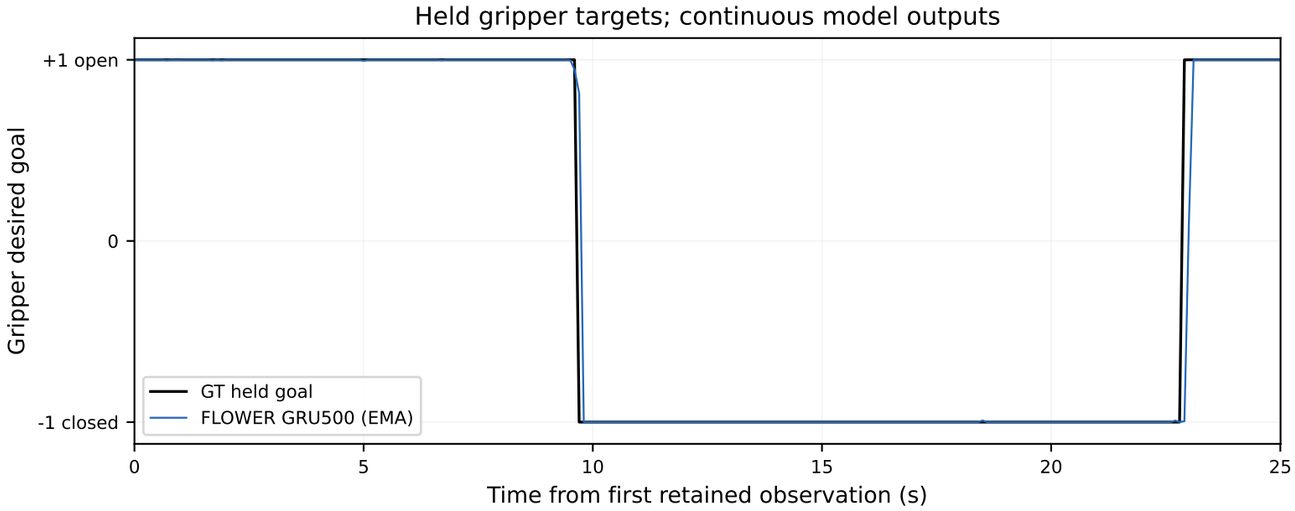
Recorded TCP and predicted command targets

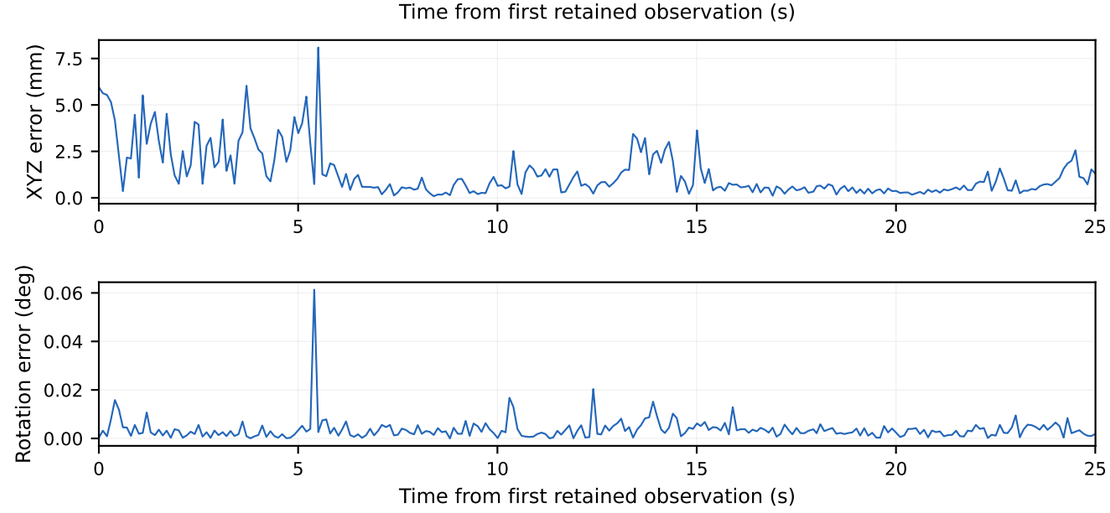


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

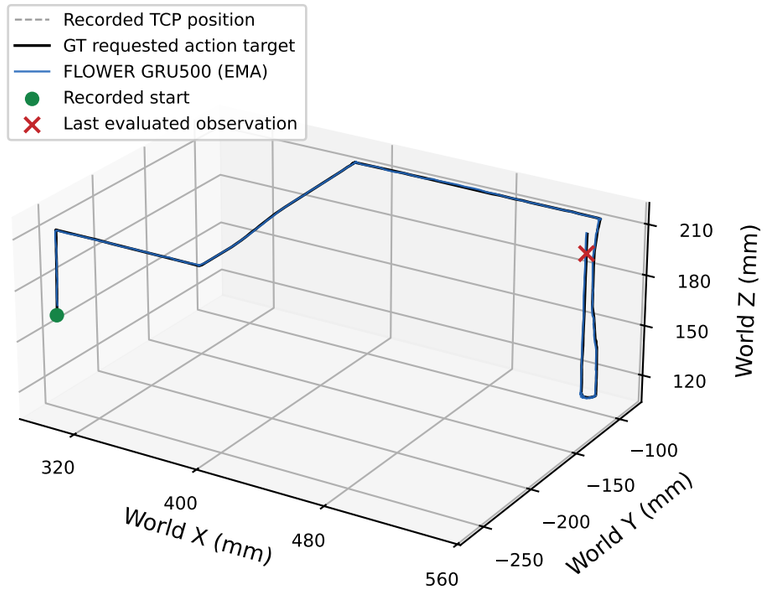
FLOWER GRU500 | barlatch p1 | episode 4/27 | RAW

1_demos_2026-10-20_18-38-46_881955.pkl

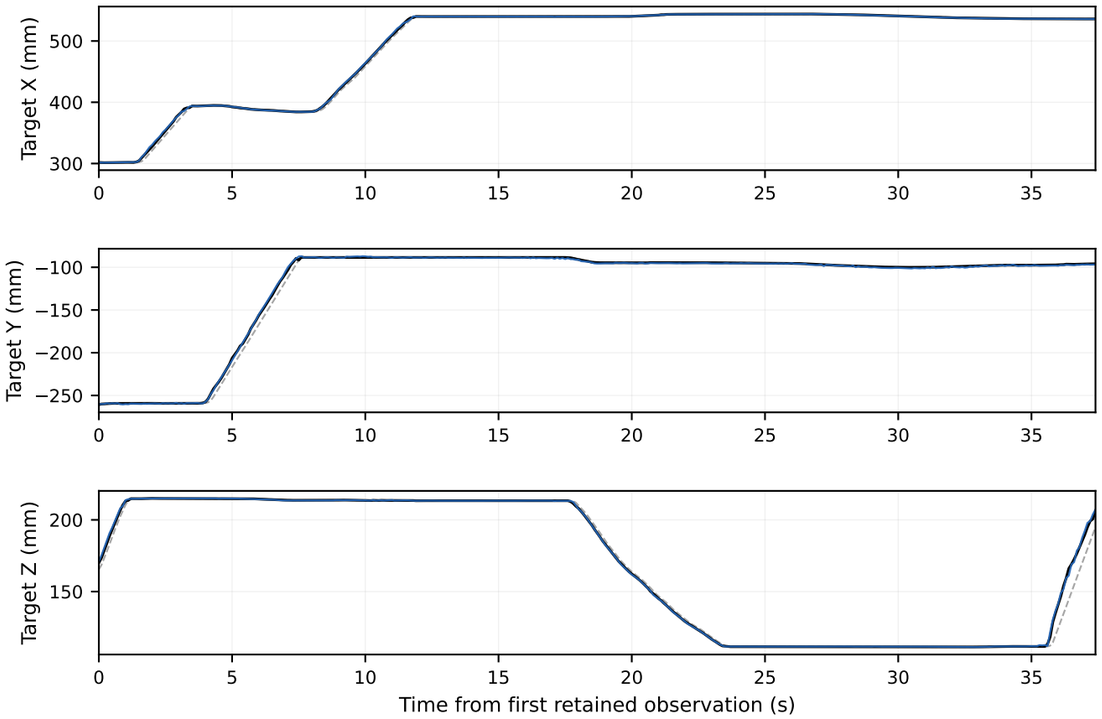
375 evaluated / 384 retained / 421 original frames | source starts at frame 37 | action rate 10 Hz

Mean XYZ target error: 0.932 mm | P95: 2.335 mm | mean rotation error: 0.135 deg | gripper sign accuracy: 98.4%

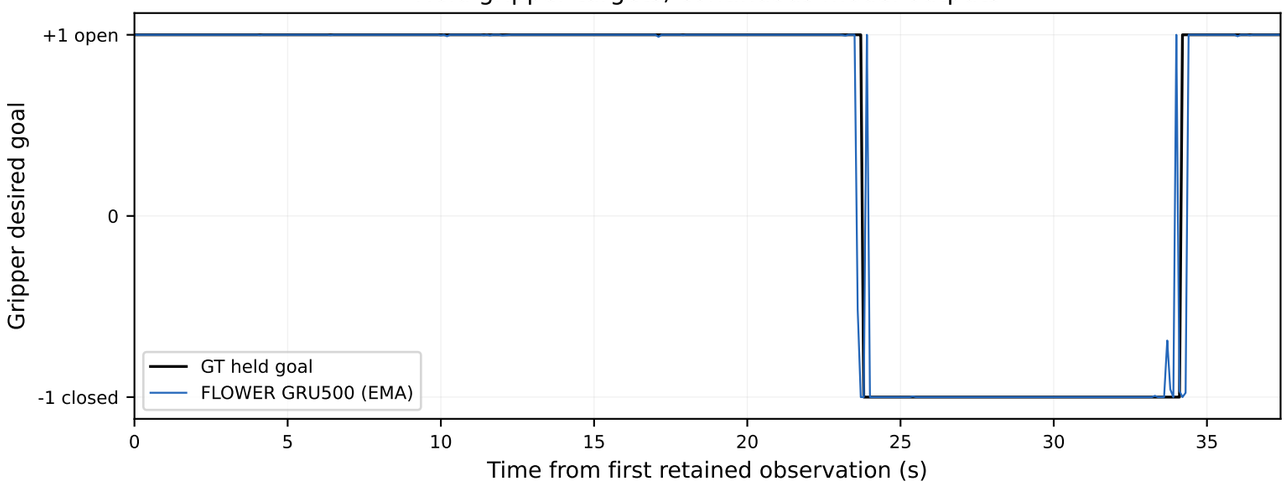
Recorded TCP and predicted command targets

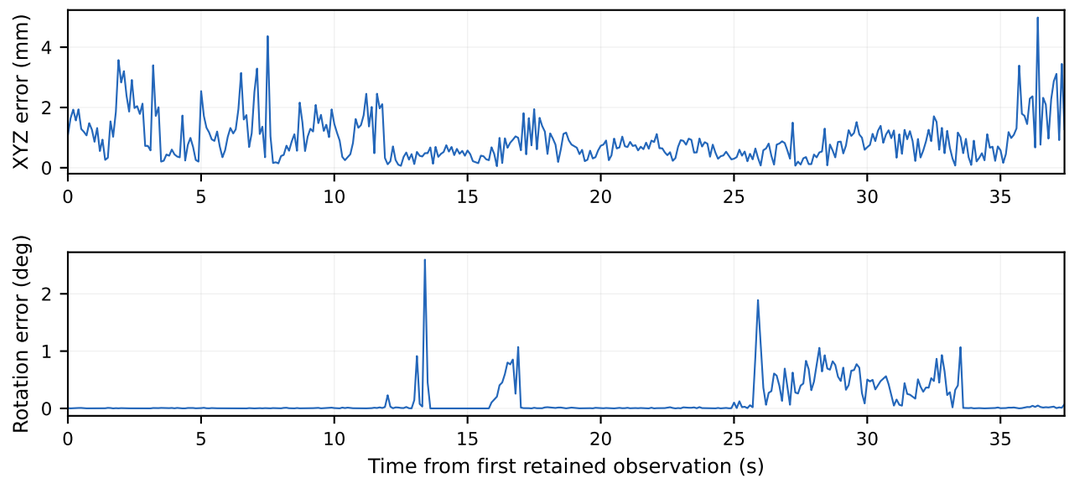


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

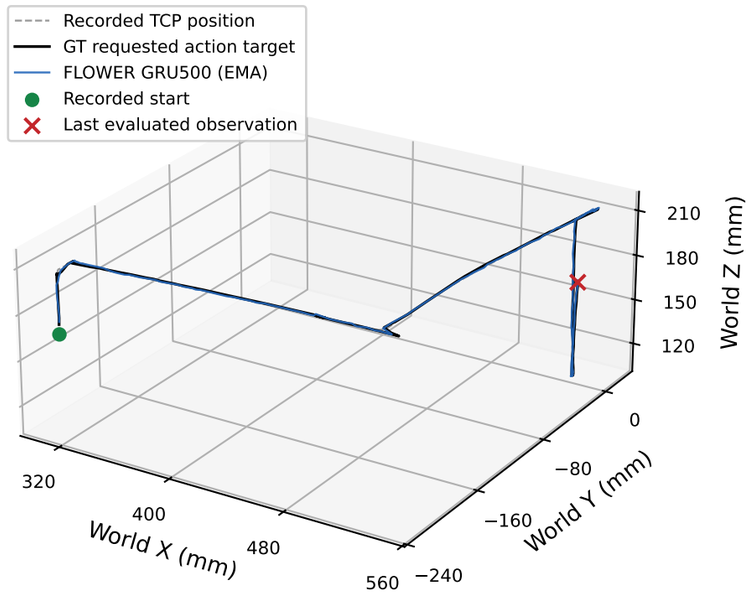
FLOWER GRU500 | barlatch p3 | episode 5/27 | RAW

1_demos_2026-10-22_20-07-16_887441.pkl

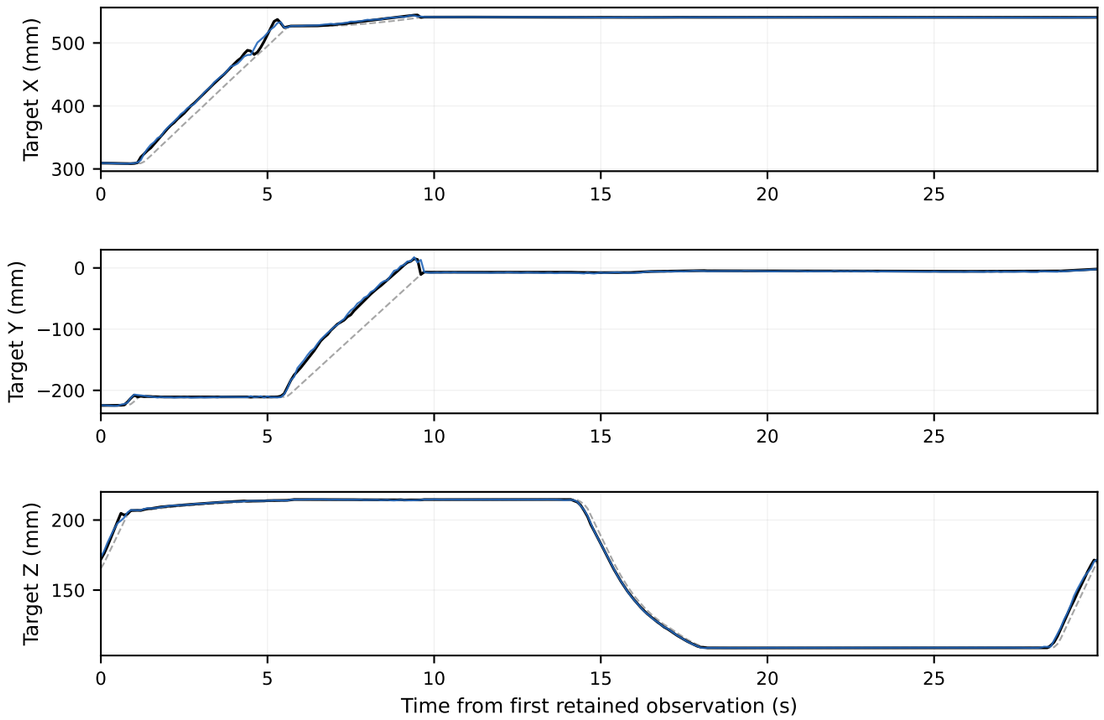
300 evaluated / 309 retained / 335 original frames | source starts at frame 26 | action rate 10 Hz

Mean XYZ target error: 1.613 mm | P95: 5.519 mm | mean rotation error: 0.147 deg | gripper sign accuracy: 99.3%

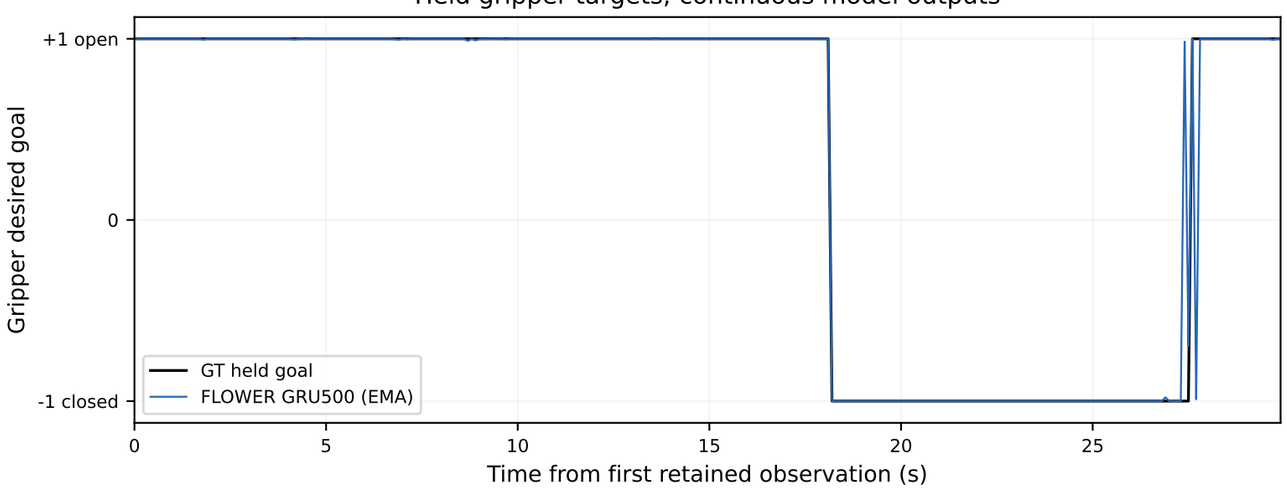
Recorded TCP and predicted command targets

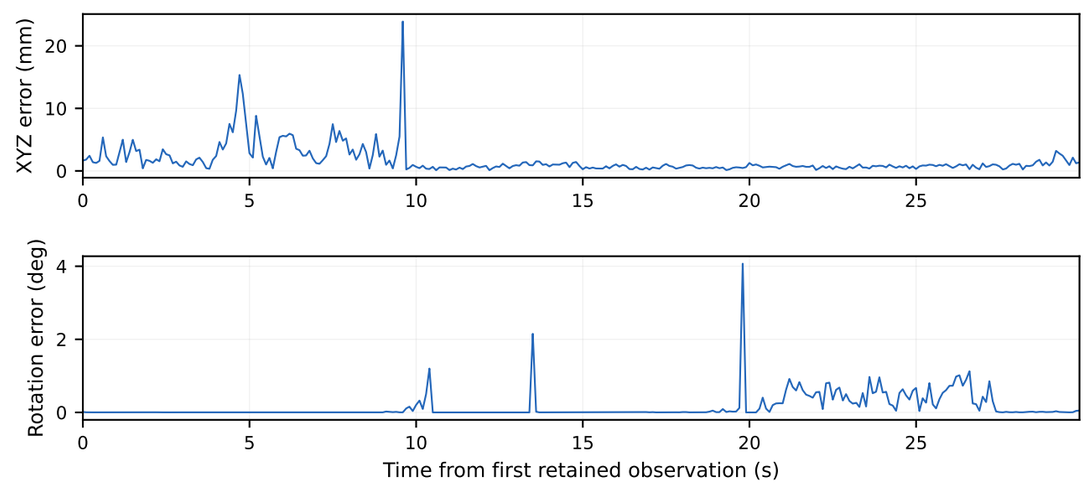


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

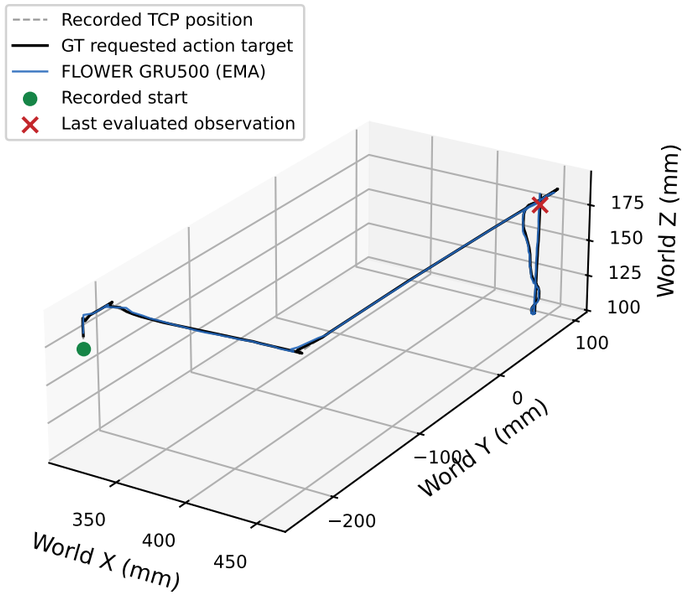
FLOWER GRU500 | barlatch p5 | episode 6/27 | RAW

1_demos_2026-10-22_21-12-13_944338.pkl

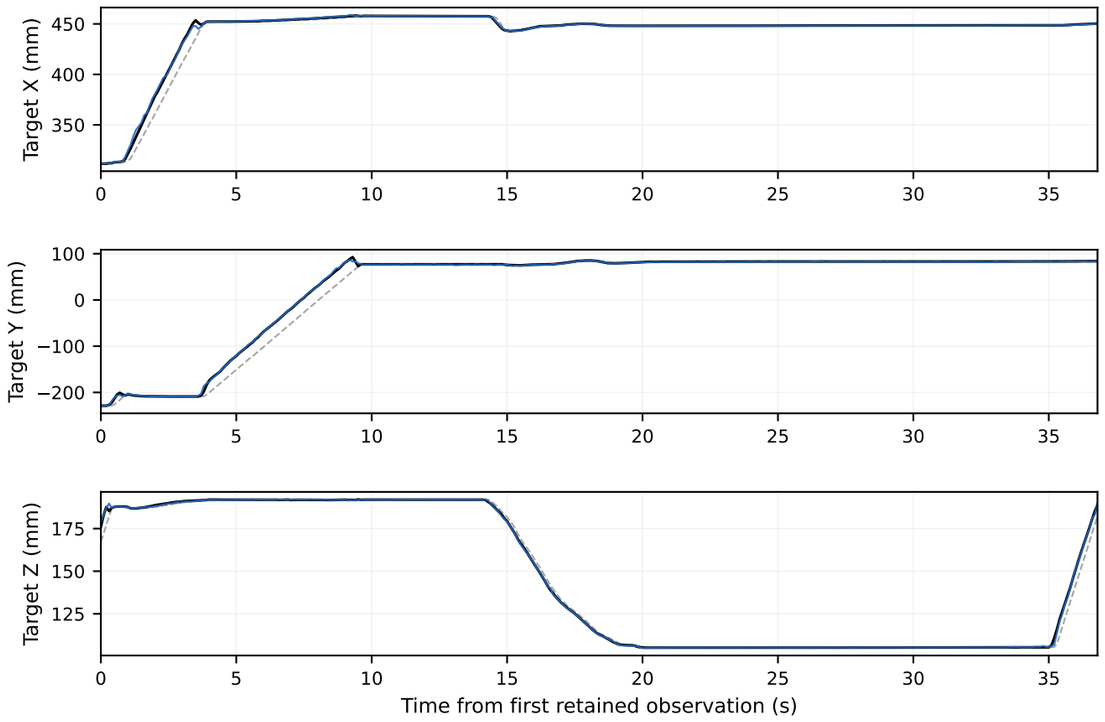
369 evaluated / 378 retained / 384 original frames | source starts at frame 6 | action rate 10 Hz

Mean XYZ target error: 1.057 mm | P95: 3.525 mm | mean rotation error: 0.137 deg | gripper sign accuracy: 98.9%

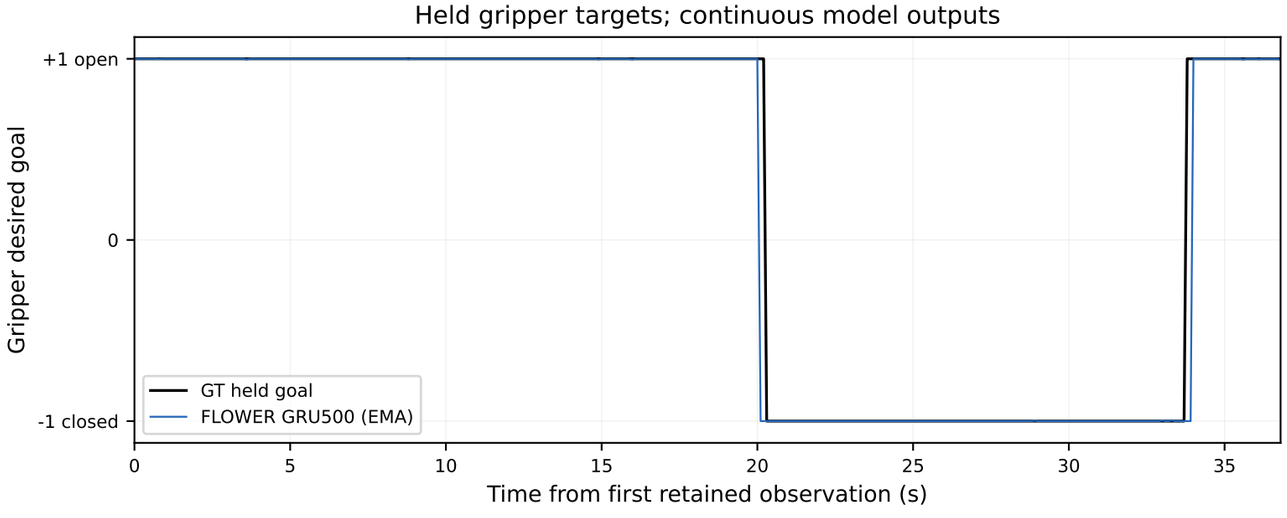
Recorded TCP and predicted command targets

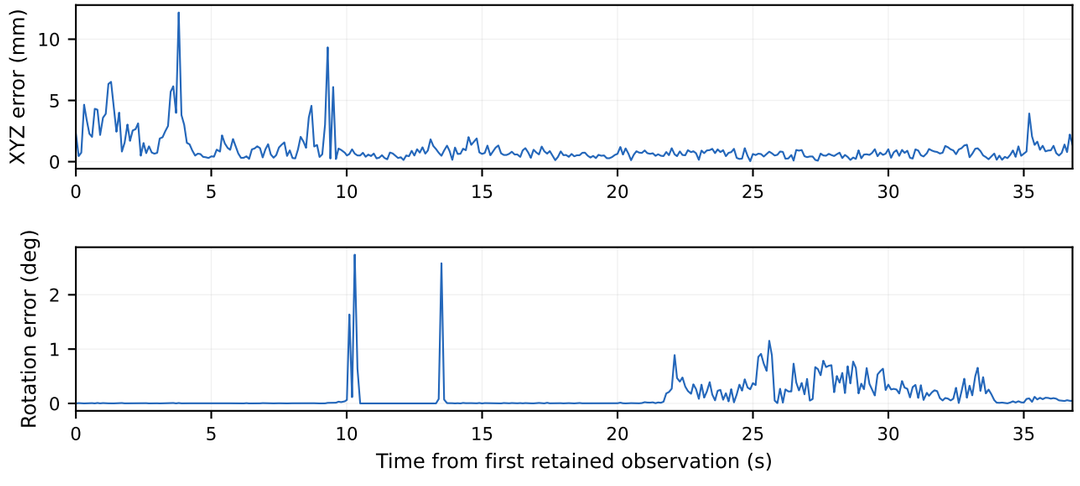


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

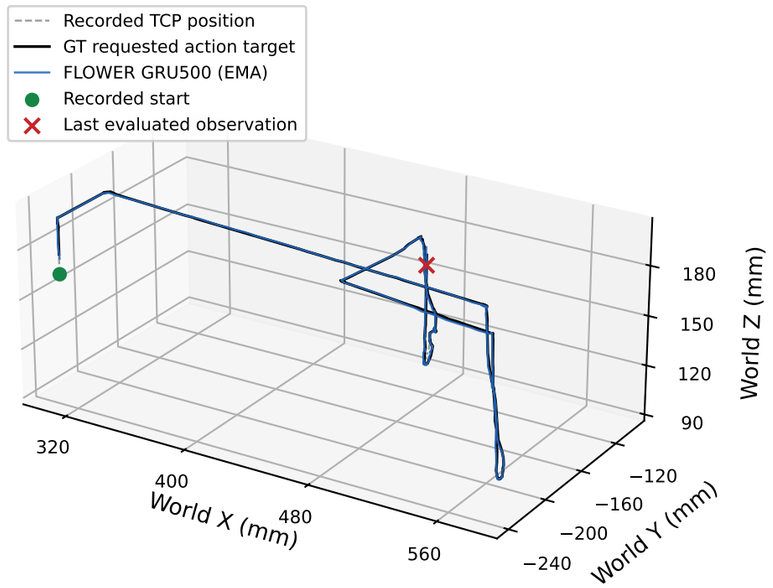
FLOWER GRU500 | bnc p1 | episode 7/27 | RAW

1_demos_2026-10-27_17-10-38_887986.pkl

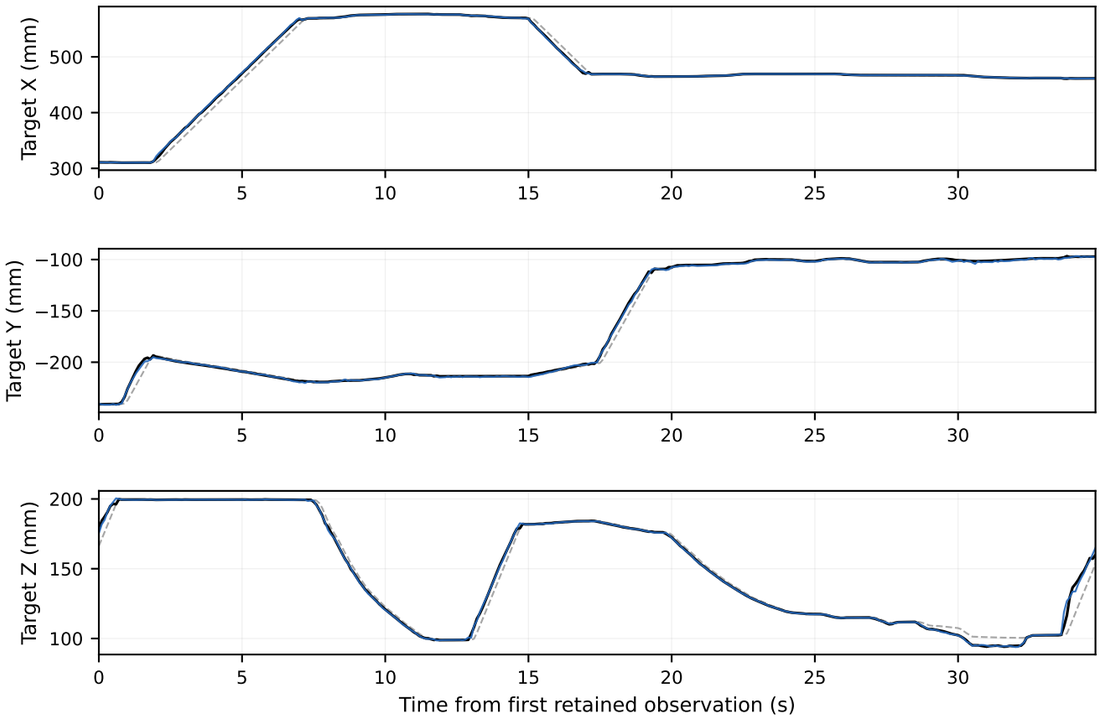
349 evaluated / 358 retained / 398 original frames | source starts at frame 40 | action rate 10 Hz

Mean XYZ target error: 1.070 mm | P95: 2.845 mm | mean rotation error: 0.065 deg | gripper sign accuracy: 98.6%

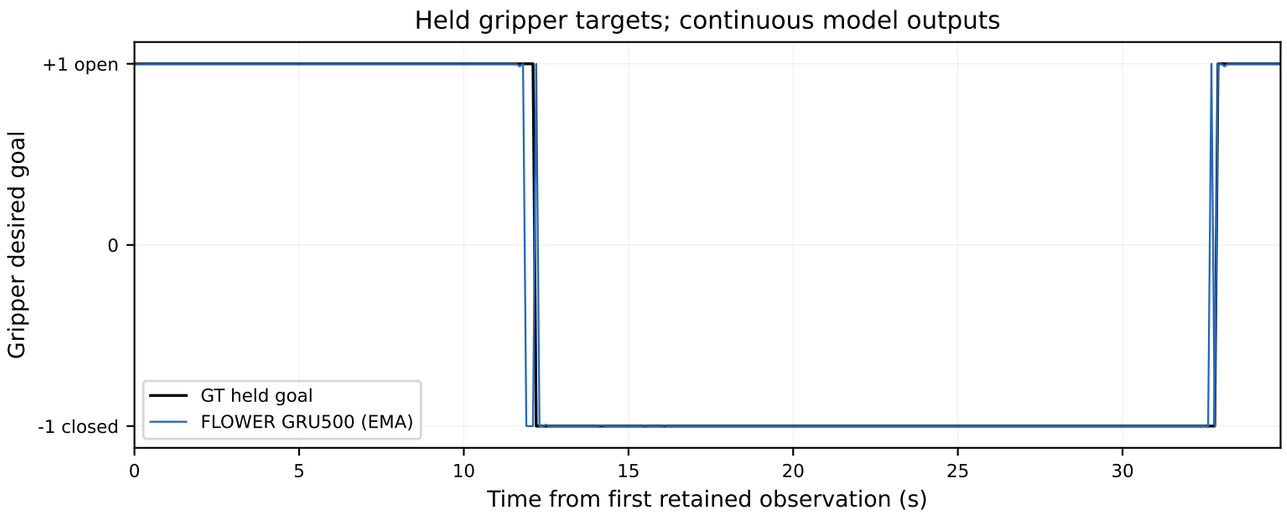
Recorded TCP and predicted command targets

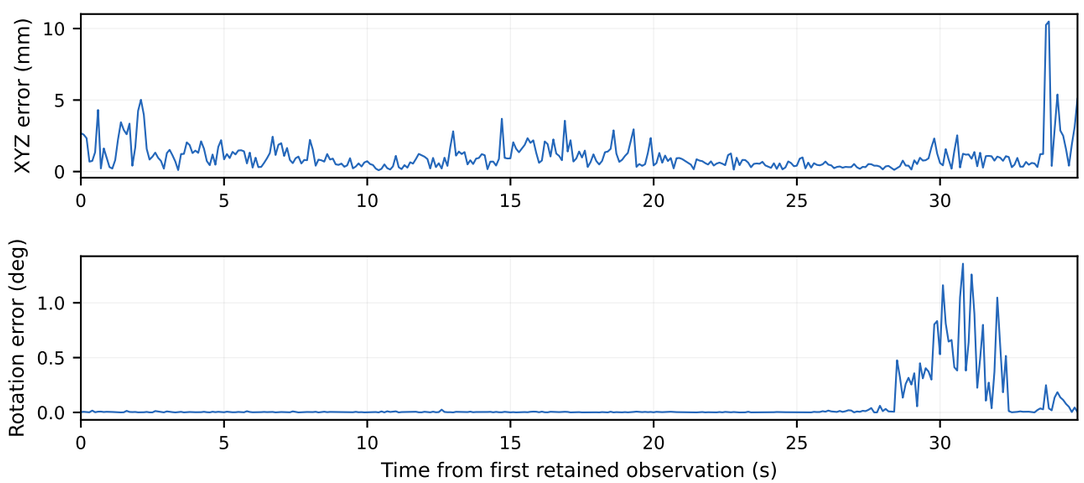


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

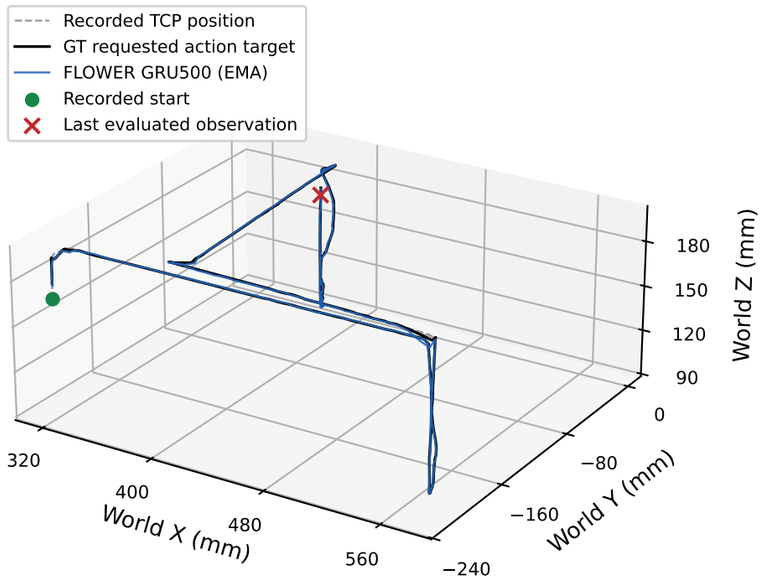
FLOWER GRU500 | bnc p3 | episode 8/27 | RAW

1_demos_2026-10-27_16-41-11_009647.pkl

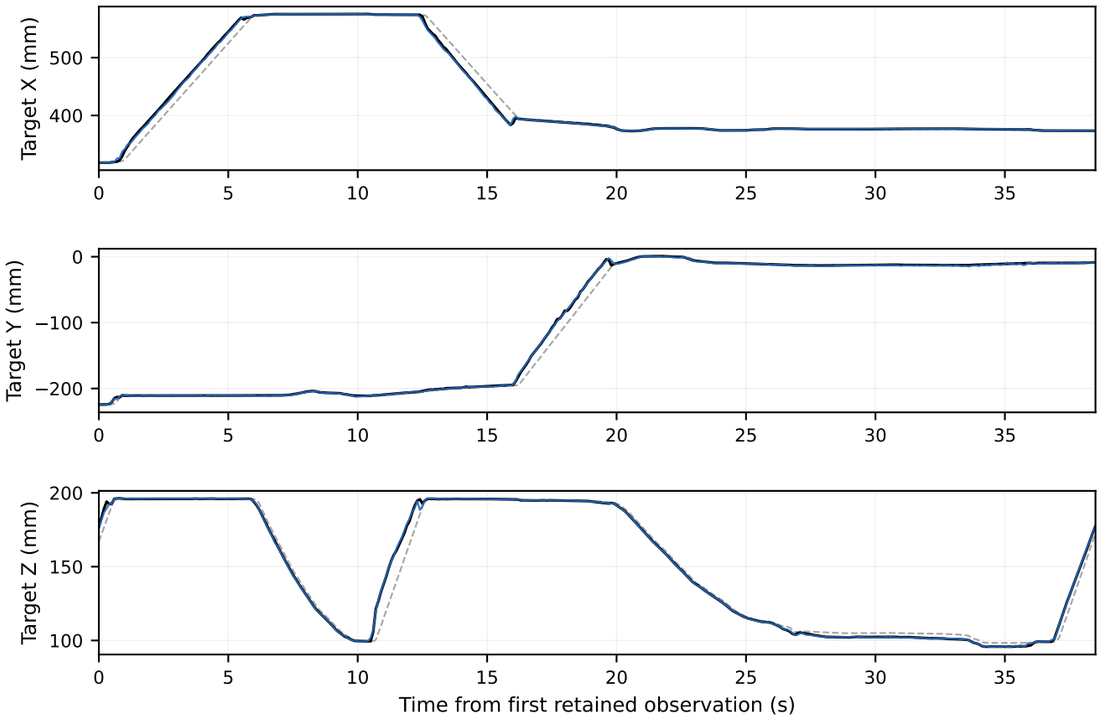
386 evaluated / 395 retained / 402 original frames | source starts at frame 7 | action rate 10 Hz

Mean XYZ target error: 1.346 mm | P95: 3.841 mm | mean rotation error: 0.039 deg | gripper sign accuracy: 99.2%

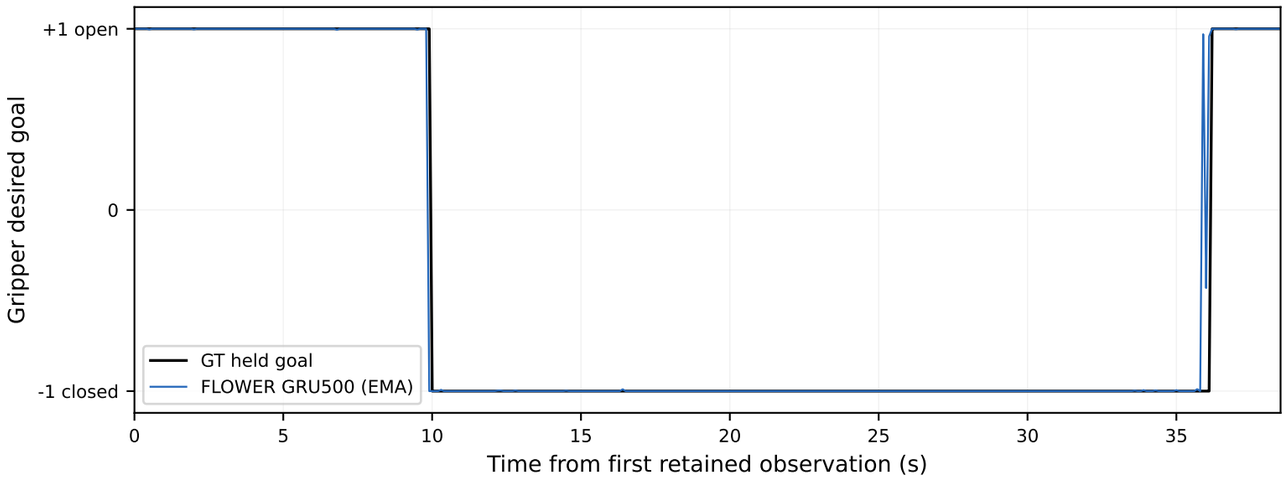
Recorded TCP and predicted command targets

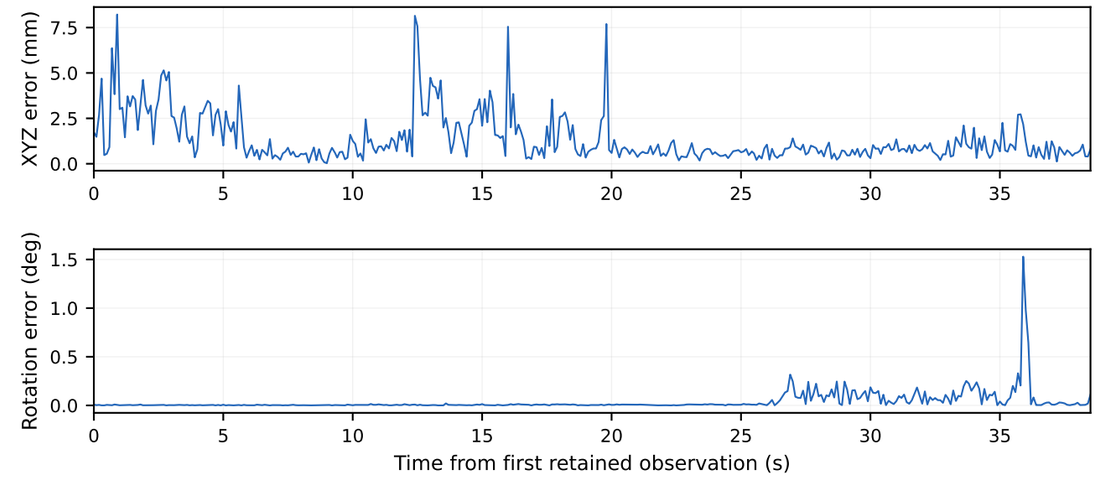


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

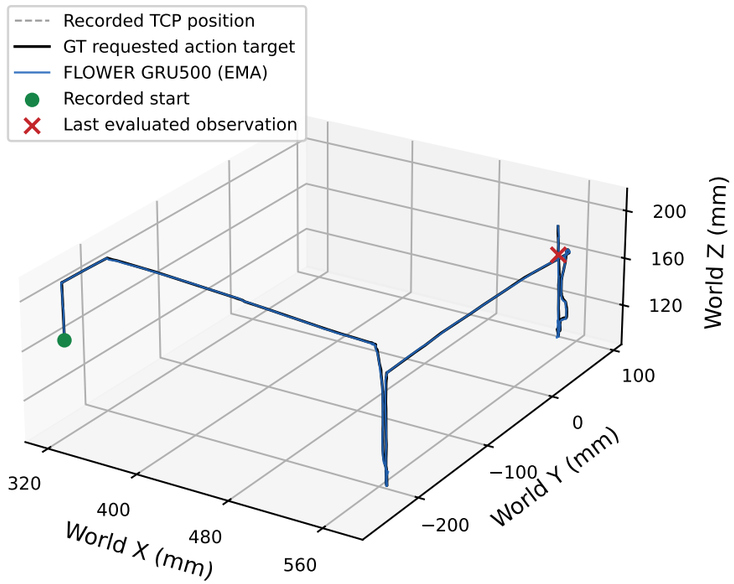
FLOWER GRU500 | bnc p5 | episode 9/27 | RAW

1_demos_2026-10-26_21-11-27_535023.pkl

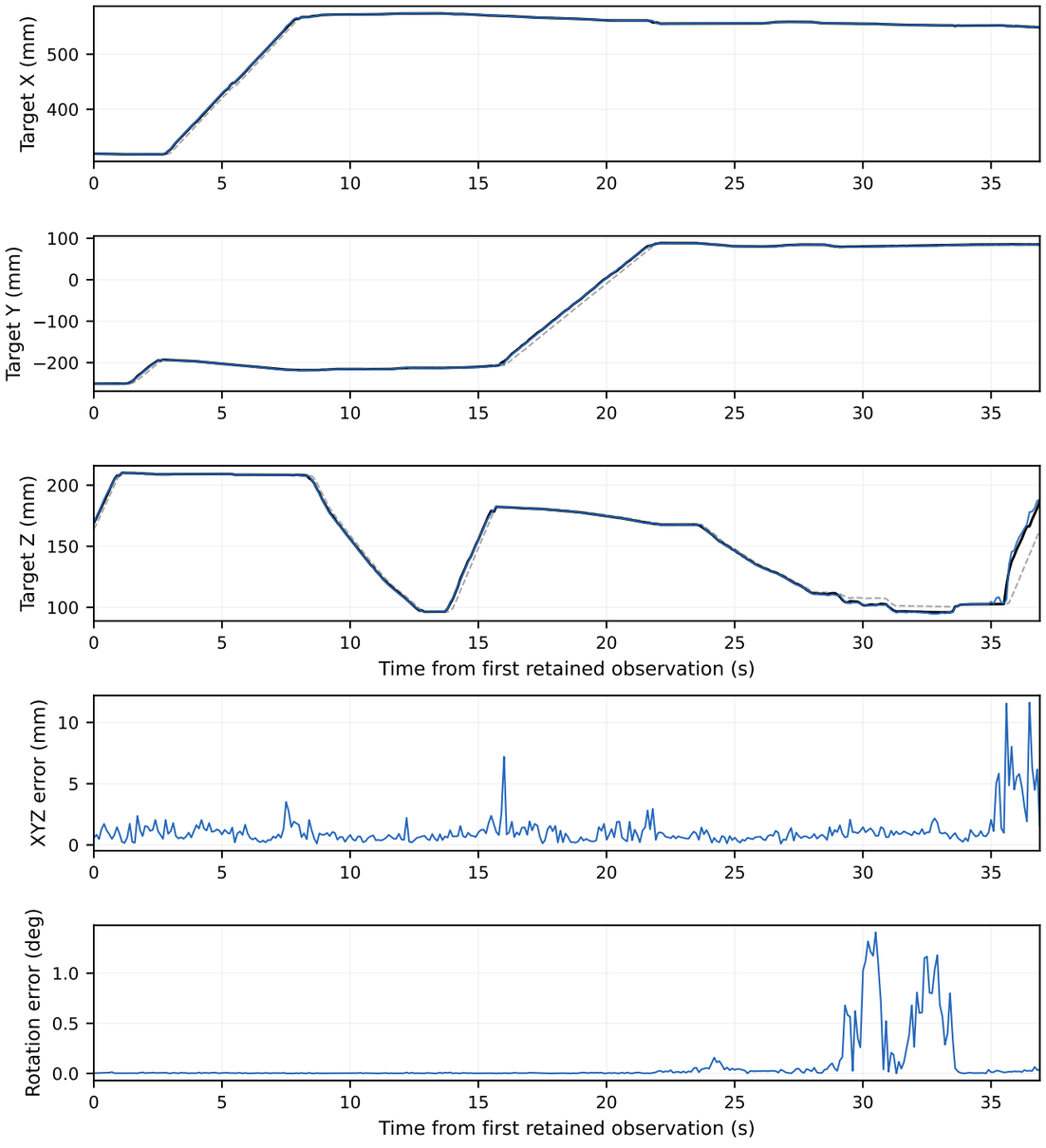
370 evaluated / 379 retained / 405 original frames | source starts at frame 26 | action rate 10 Hz

Mean XYZ target error: 1.121 mm | P95: 2.661 mm | mean rotation error: 0.082 deg | gripper sign accuracy: 99.2%

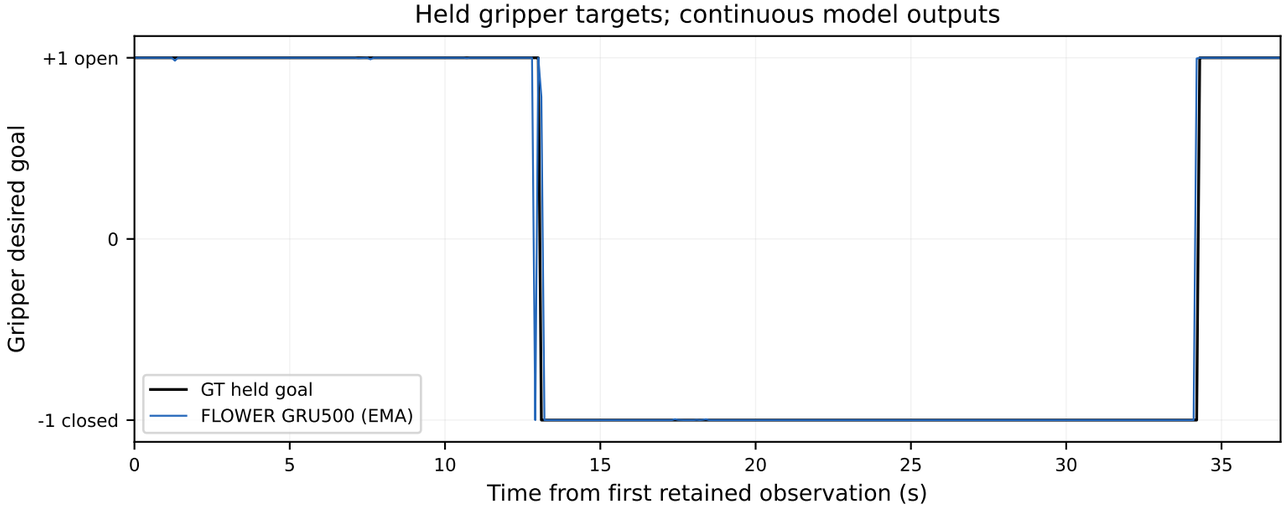
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

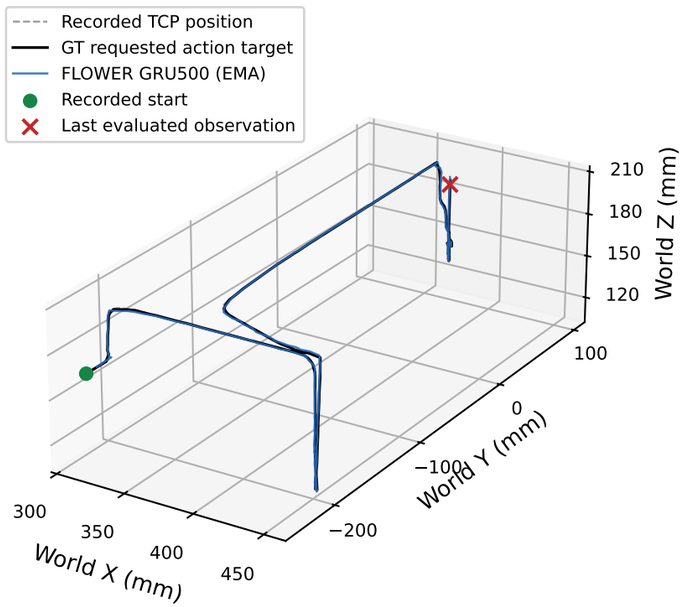
FLOWER GRU500 | hdmi p1 | episode 10/27 | RAW

1_demos_2026-10-19_17-49-12_080506.pkl

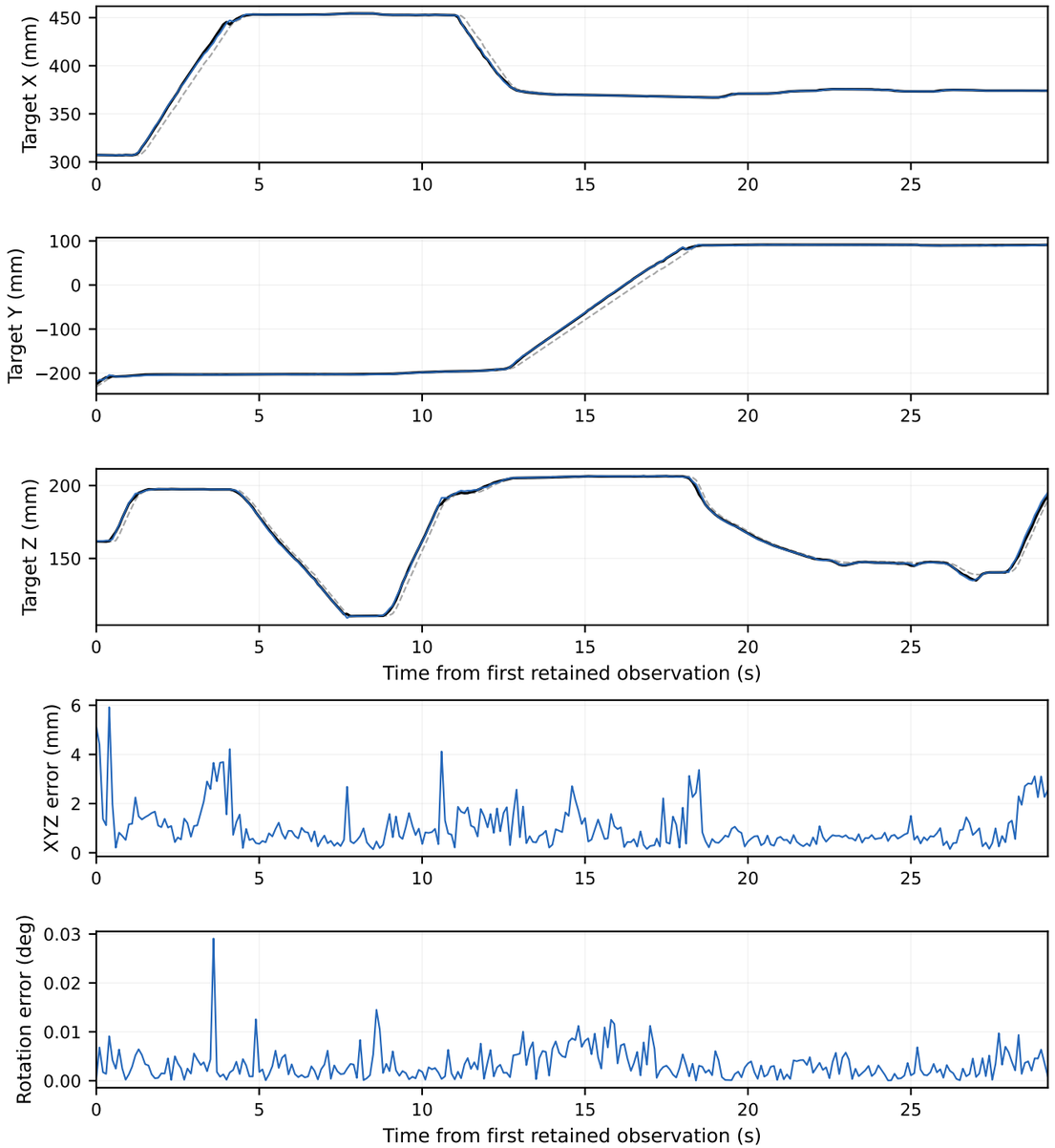
293 evaluated / 302 retained / 310 original frames | source starts at frame 8 | action rate 10 Hz

Mean XYZ target error: 1.026 mm | P95: 2.806 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 99.3%

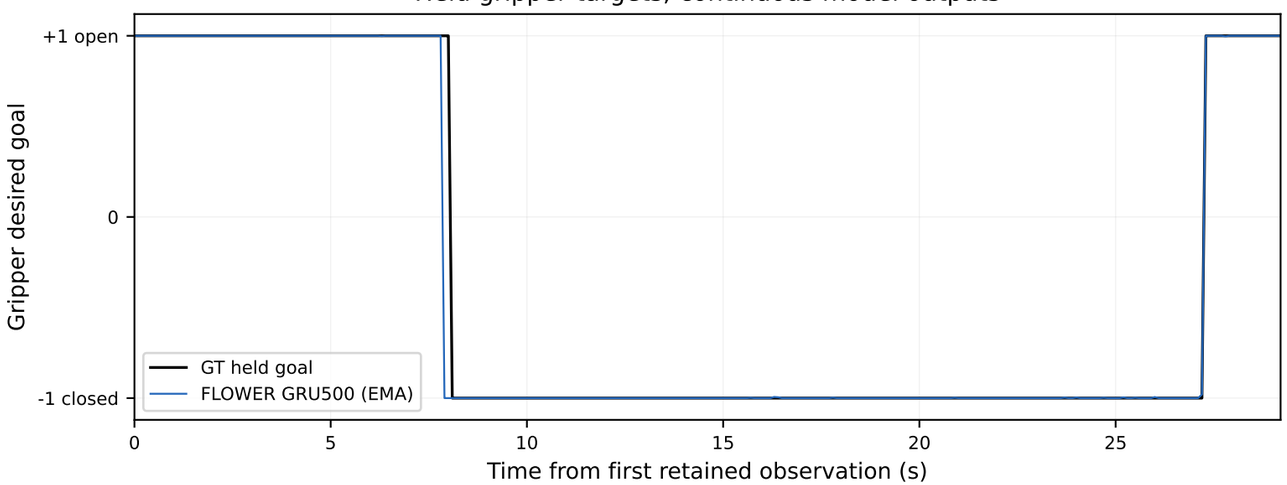
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

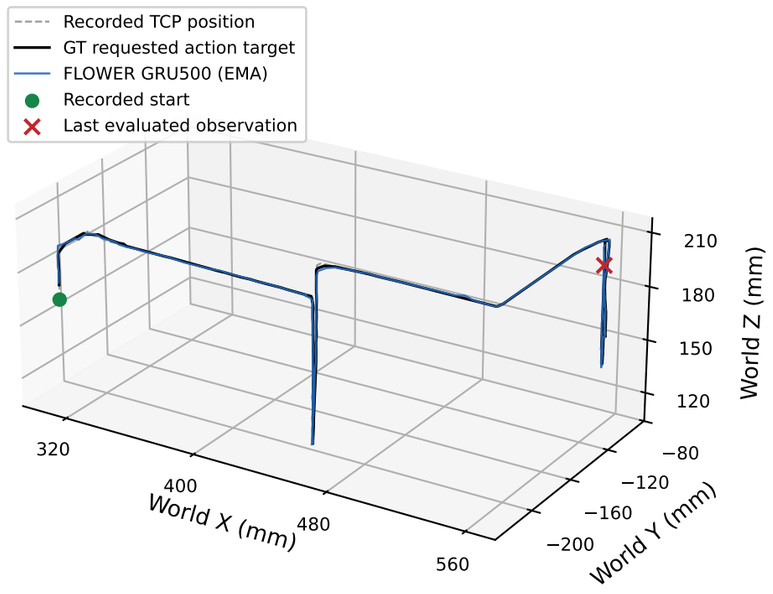
FLOWER GRU500 | hdmi p3 | episode 11/27 | RAW

1_demos_2026-10-21_20-09-03_529204.pkl

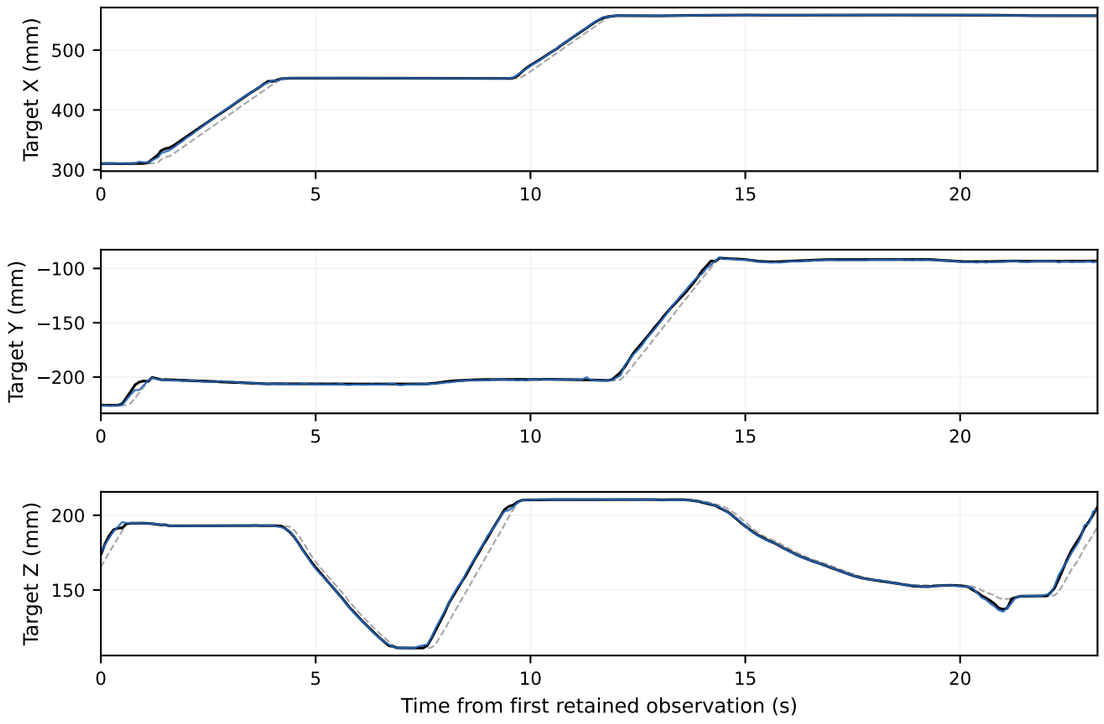
233 evaluated / 242 retained / 261 original frames | source starts at frame 19 | action rate 10 Hz

Mean XYZ target error: 1.104 mm | P95: 2.951 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 100.0%

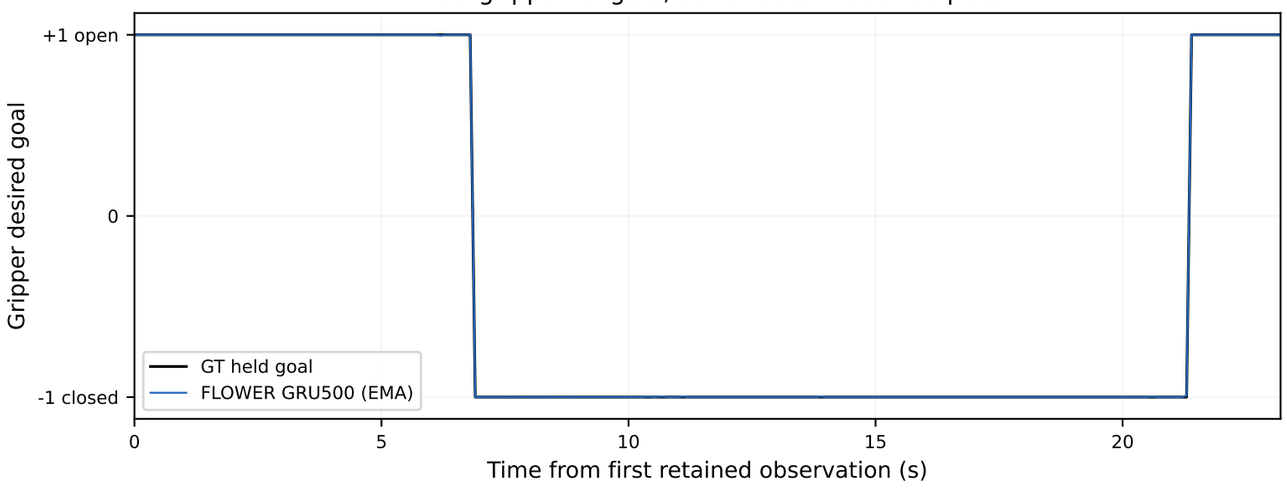
Recorded TCP and predicted command targets

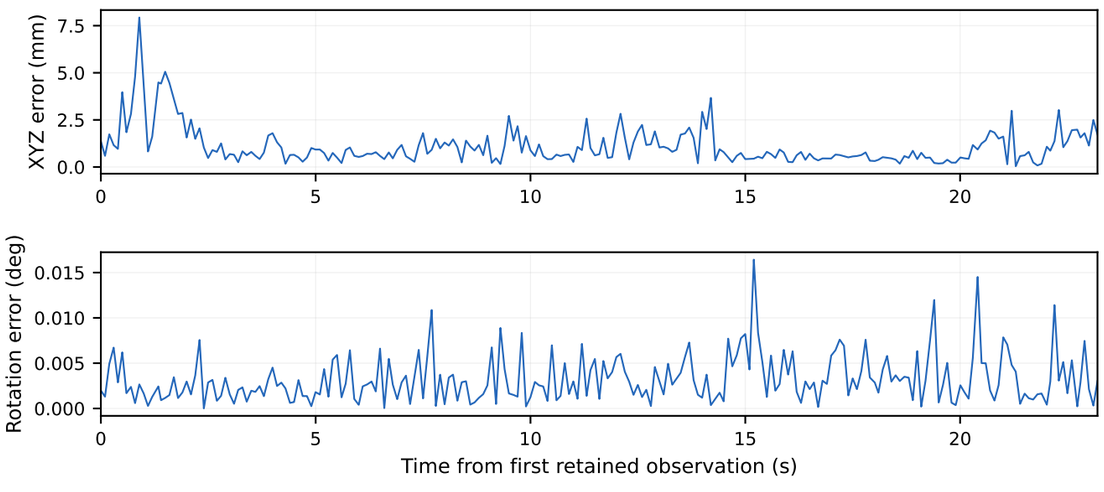


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

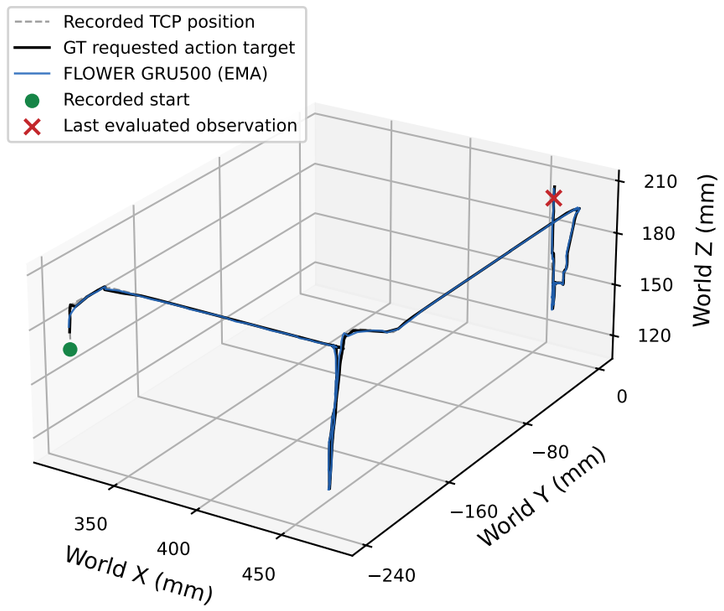
FLOWER GRU500 | hdmi p5 | episode 12/27 | RAW

1_demos_2026-10-22_20-59-43_175476.pkl

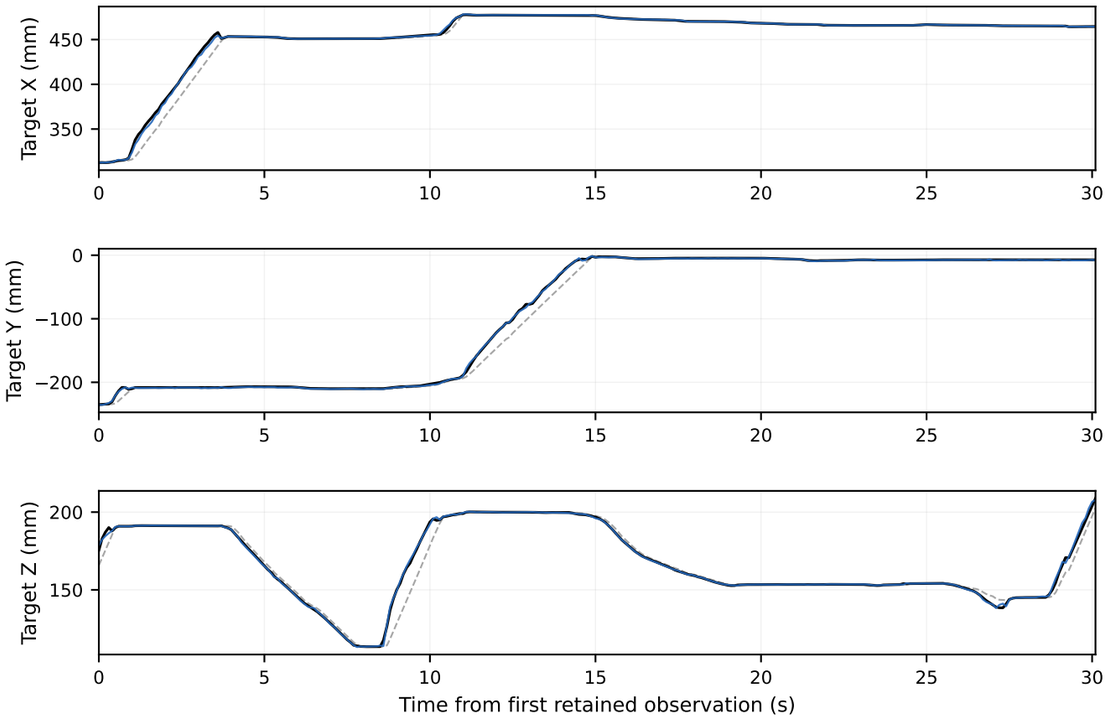
302 evaluated / 311 retained / 319 original frames | source starts at frame 8 | action rate 10 Hz

Mean XYZ target error: 1.099 mm | P95: 3.758 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 100.0%

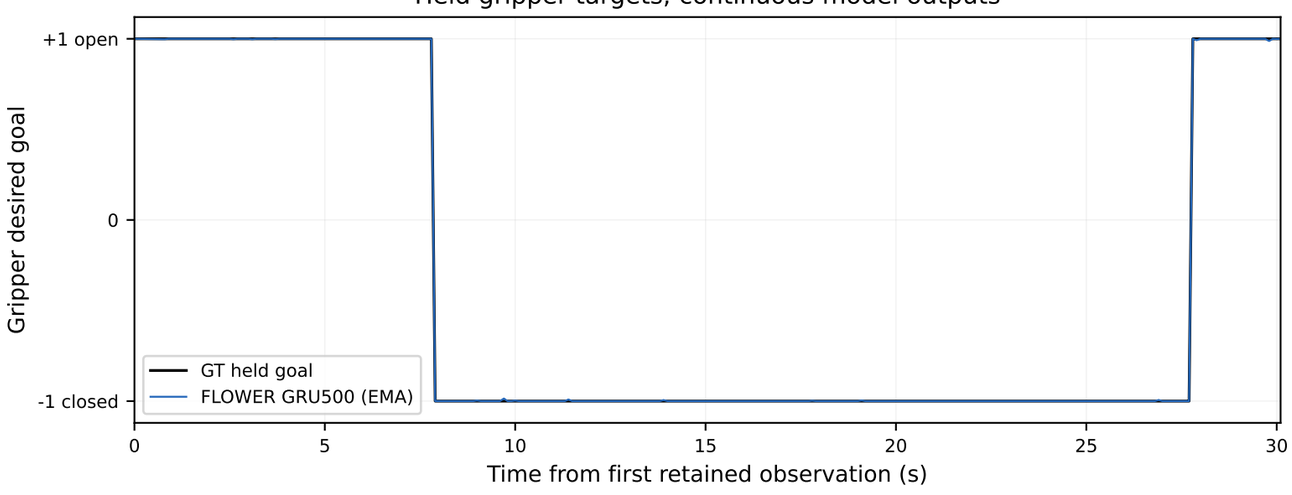
Recorded TCP and predicted command targets

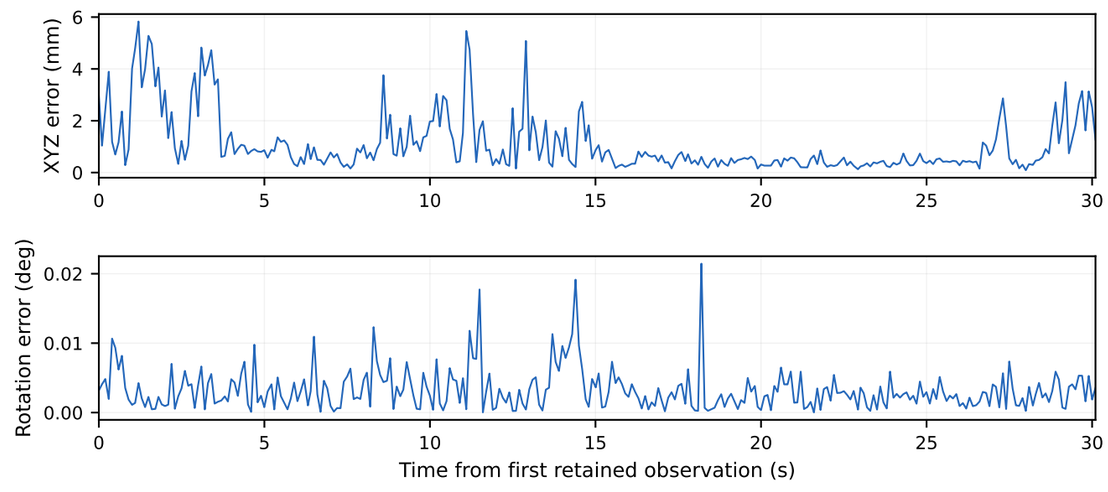


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

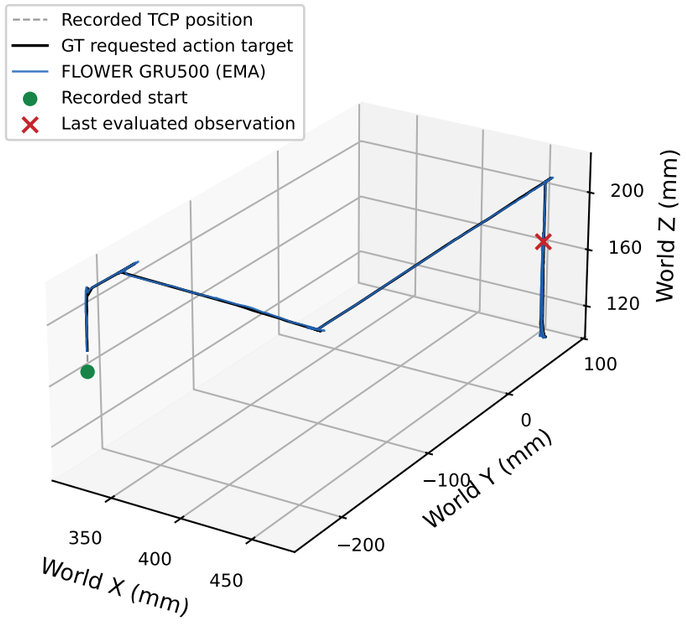
FLOWER GRU500 | knob p1 | episode 13/27 | RAW

1_demos_2026-10-20_18-26-33_442361.pkl

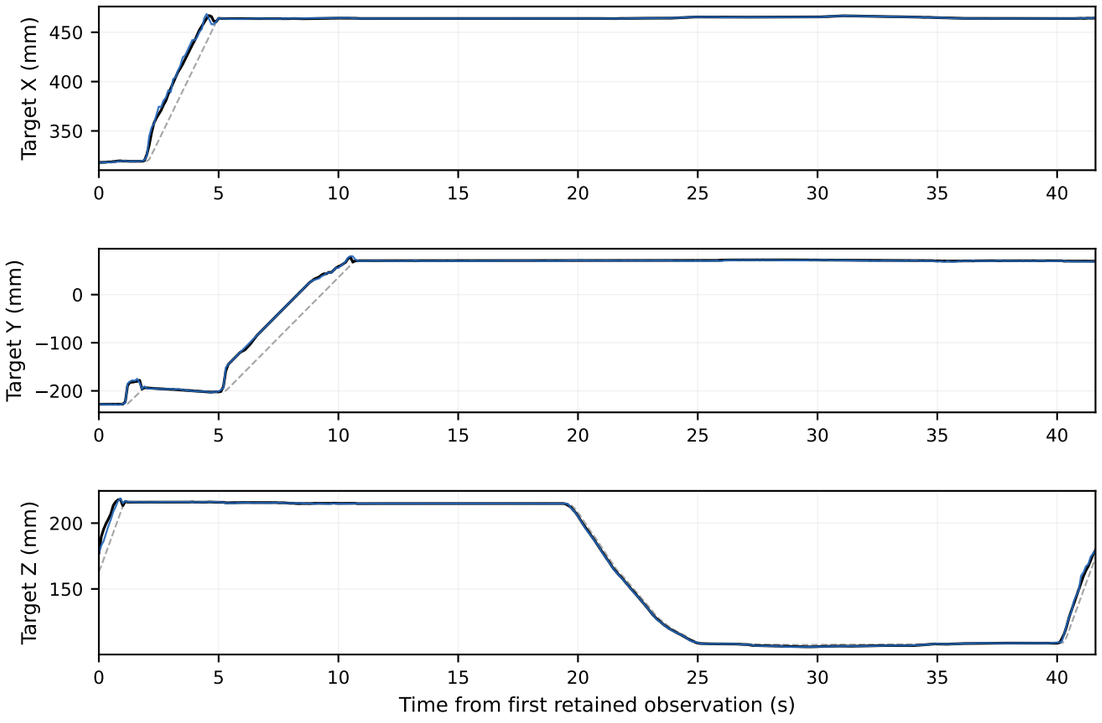
417 evaluated / 426 retained / 437 original frames | source starts at frame 11 | action rate 10 Hz

Mean XYZ target error: 1.123 mm | P95: 4.404 mm | mean rotation error: 0.083 deg | gripper sign accuracy: 99.3%

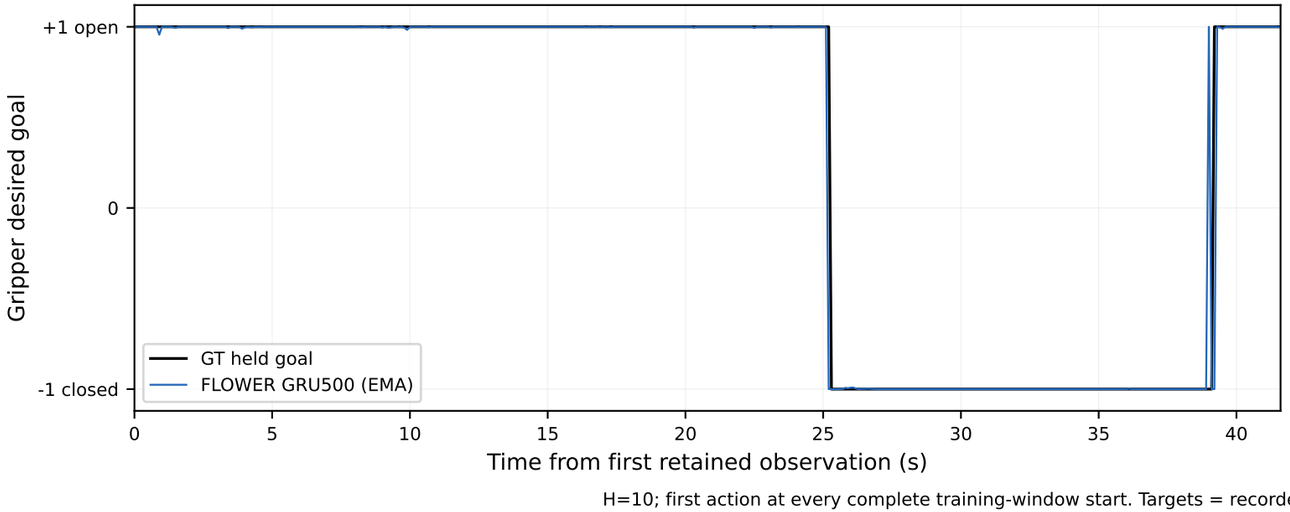
Recorded TCP and predicted command targets

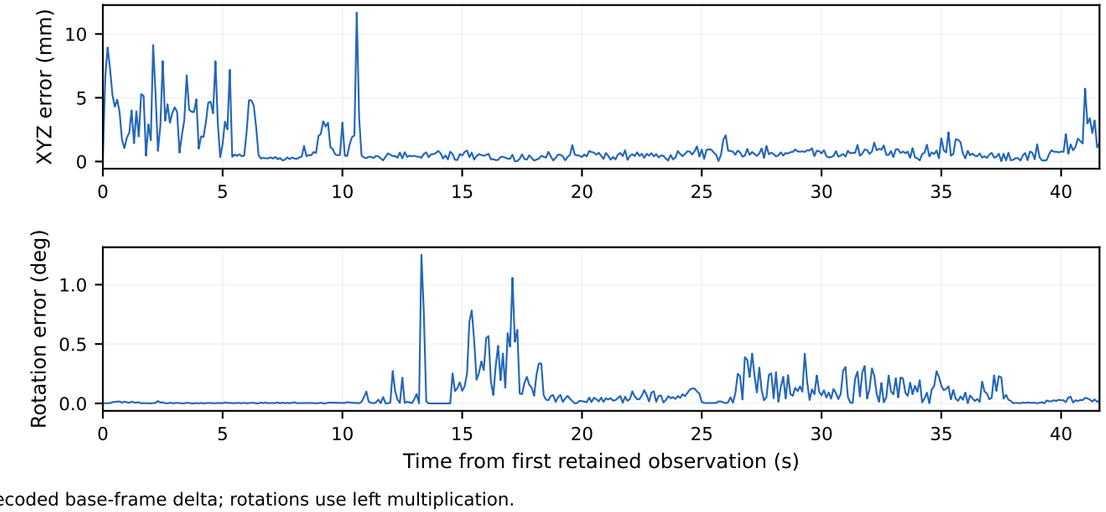


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

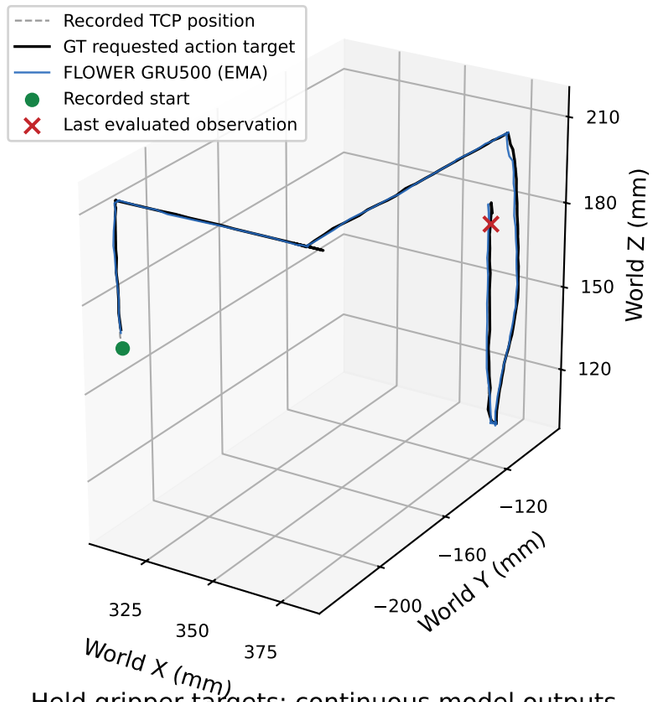
FLOWER GRU500 | knob p3 | episode 14/27 | RAW

1_demos_2026-10-21_19-14-29_954801.pkl

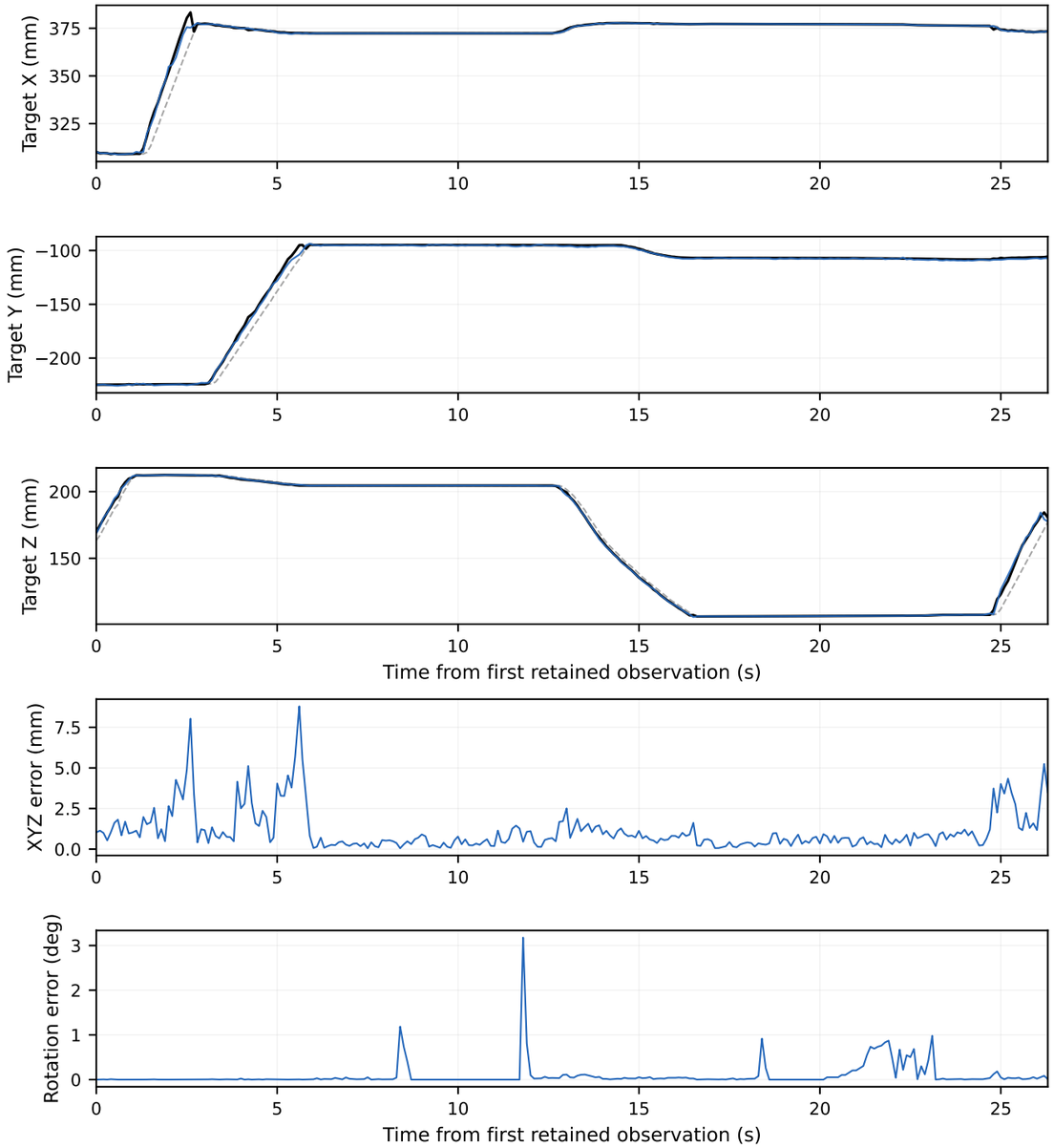
264 evaluated / 273 retained / 322 original frames | source starts at frame 49 | action rate 10 Hz

Mean XYZ target error: 1.112 mm | P95: 3.780 mm | mean rotation error: 0.087 deg | gripper sign accuracy: 98.9%

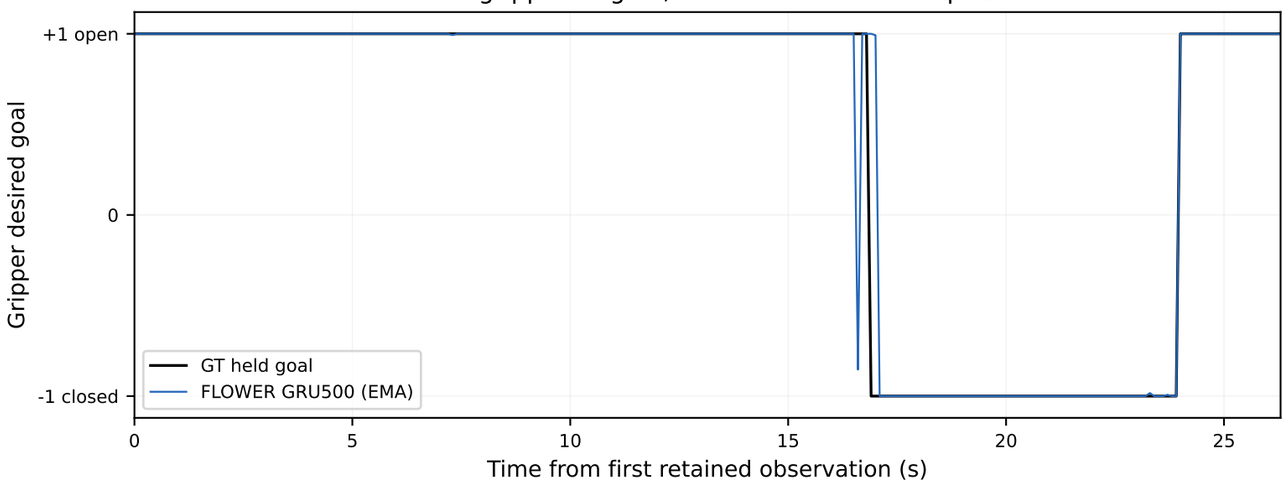
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

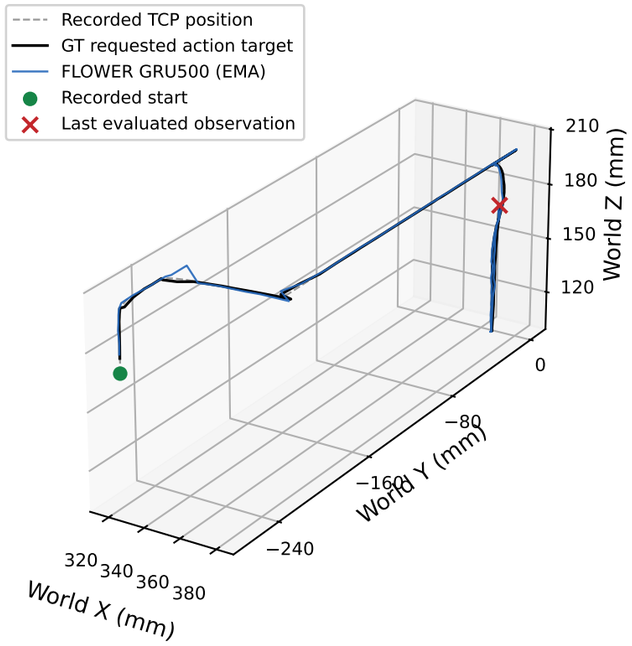
FLOWER GRU500 | knob p5 | episode 15/27 | RAW

1_demos_2026-10-22_20-34-40_592540.pkl

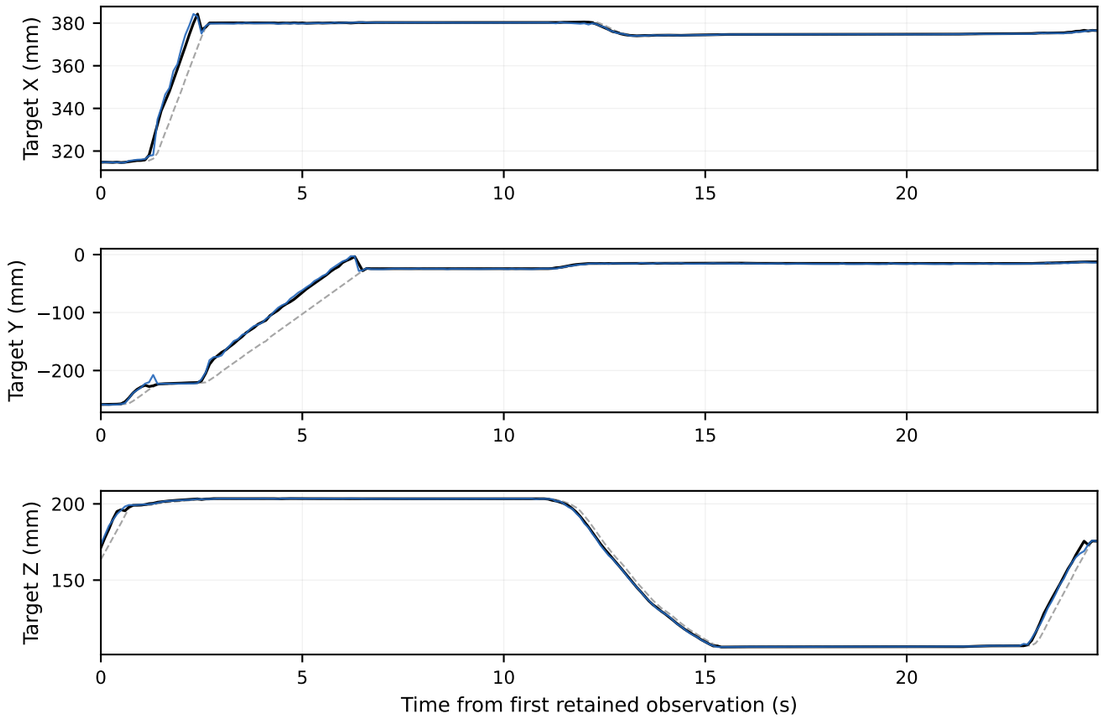
248 evaluated / 257 retained / 262 original frames | source starts at frame 5 | action rate 10 Hz

Mean XYZ target error: 1.430 mm | P95: 4.540 mm | mean rotation error: 0.130 deg | gripper sign accuracy: 99.2%

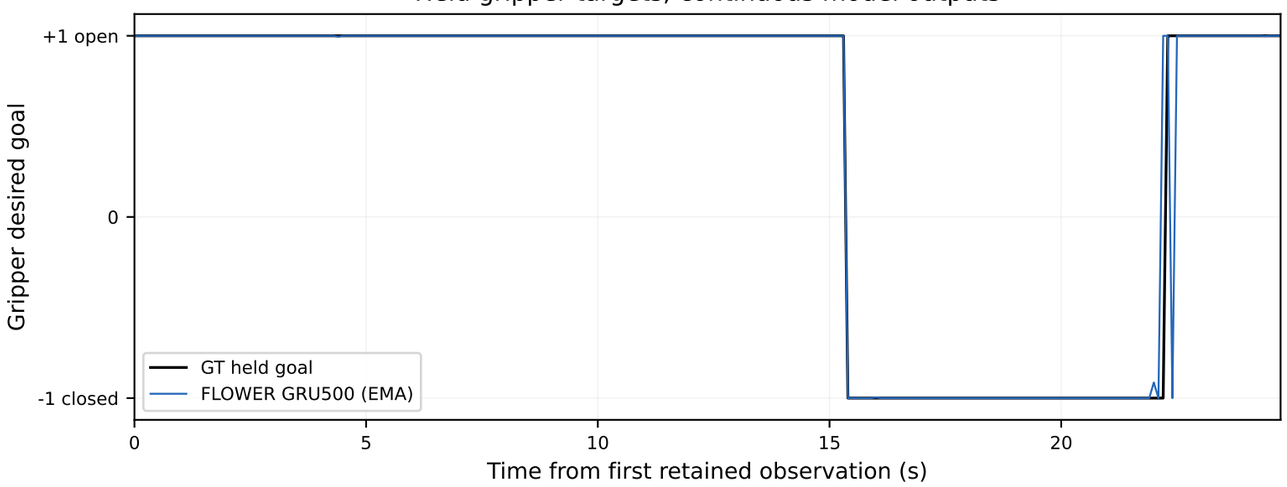
Recorded TCP and predicted command targets

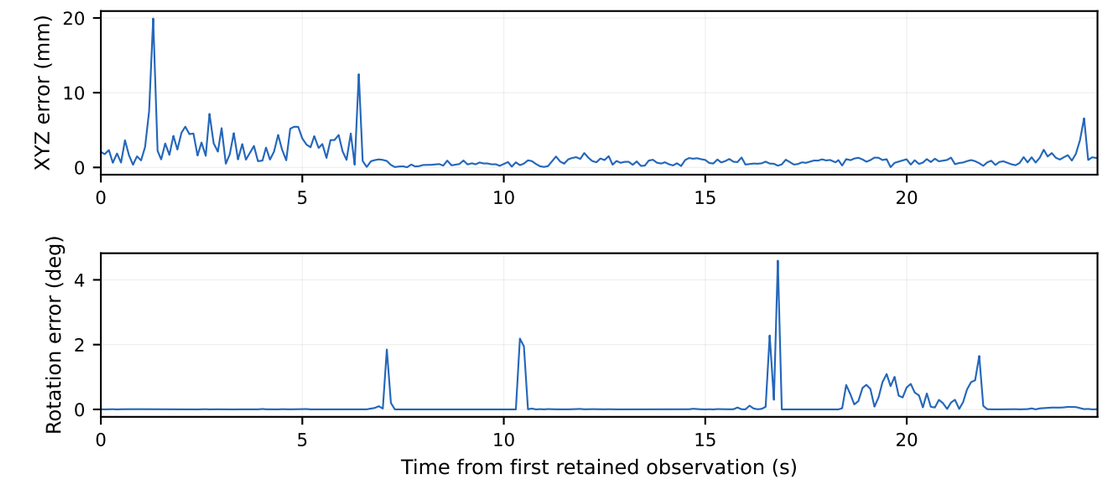


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

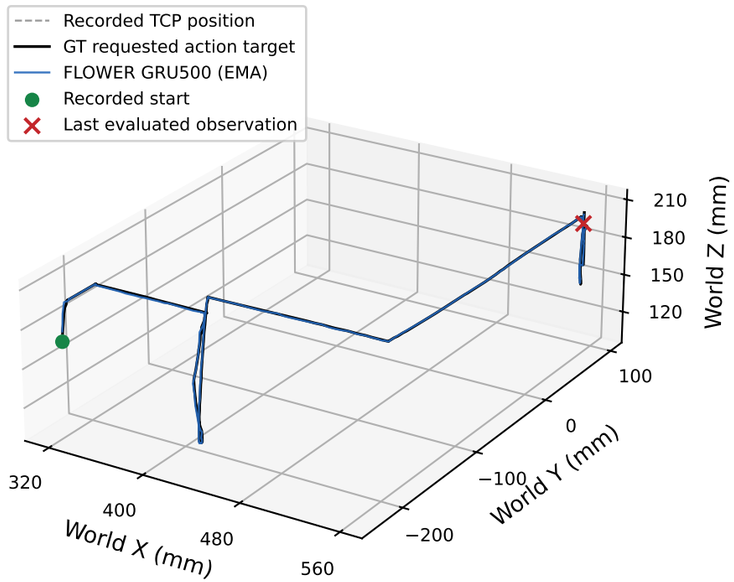
FLOWER GRU500 | lan p1 | episode 16/27 | RAW

1_demos_2026-10-20_19-16-57_244675.pkl

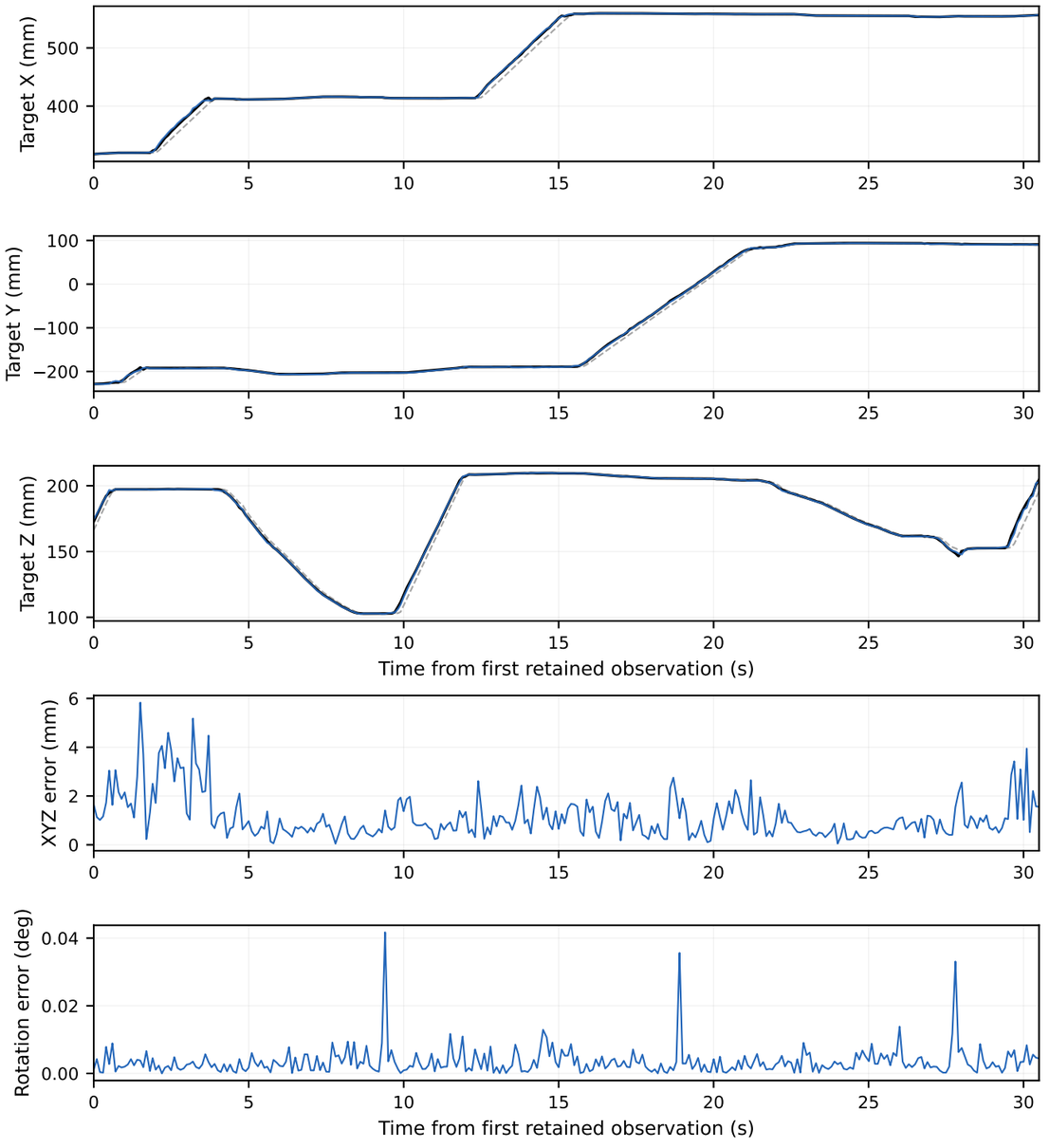
306 evaluated / 315 retained / 340 original frames | source starts at frame 25 | action rate 10 Hz

Mean XYZ target error: 1.139 mm | P95: 3.091 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 99.0%

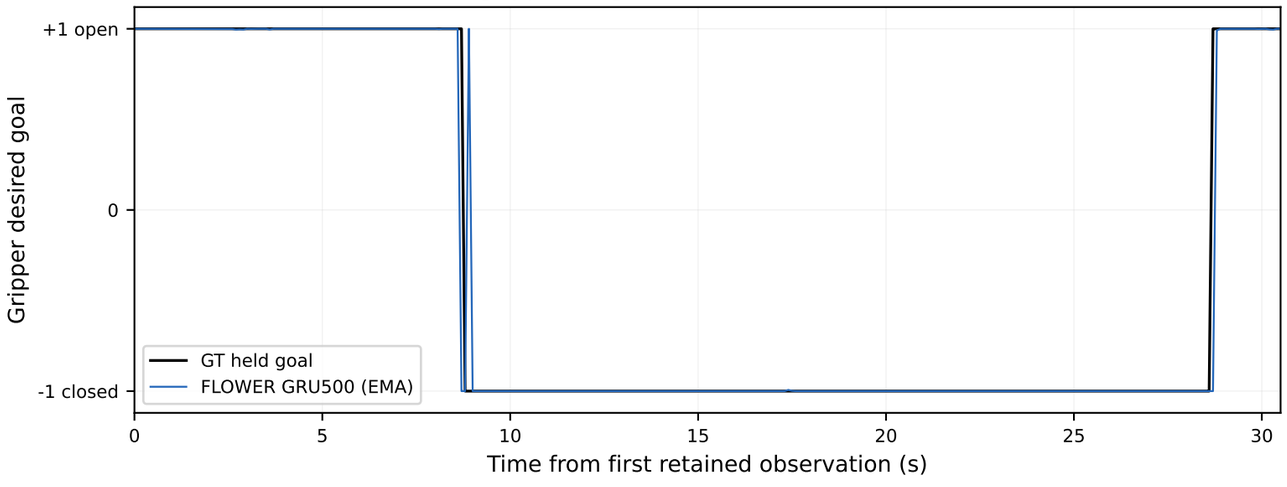
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

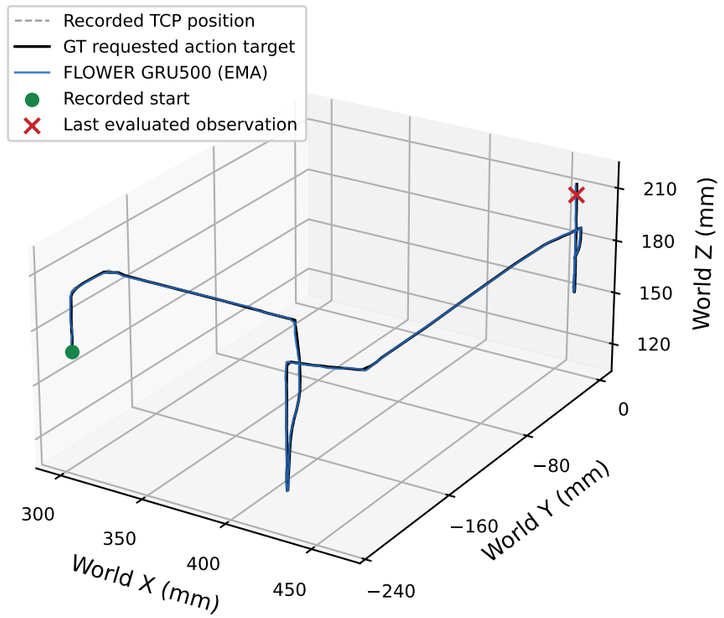
FLOWER GRU500 | lan p3 | episode 17/27 | RAW

1_demos_2026-10-21_19-58-42_512708.pkl

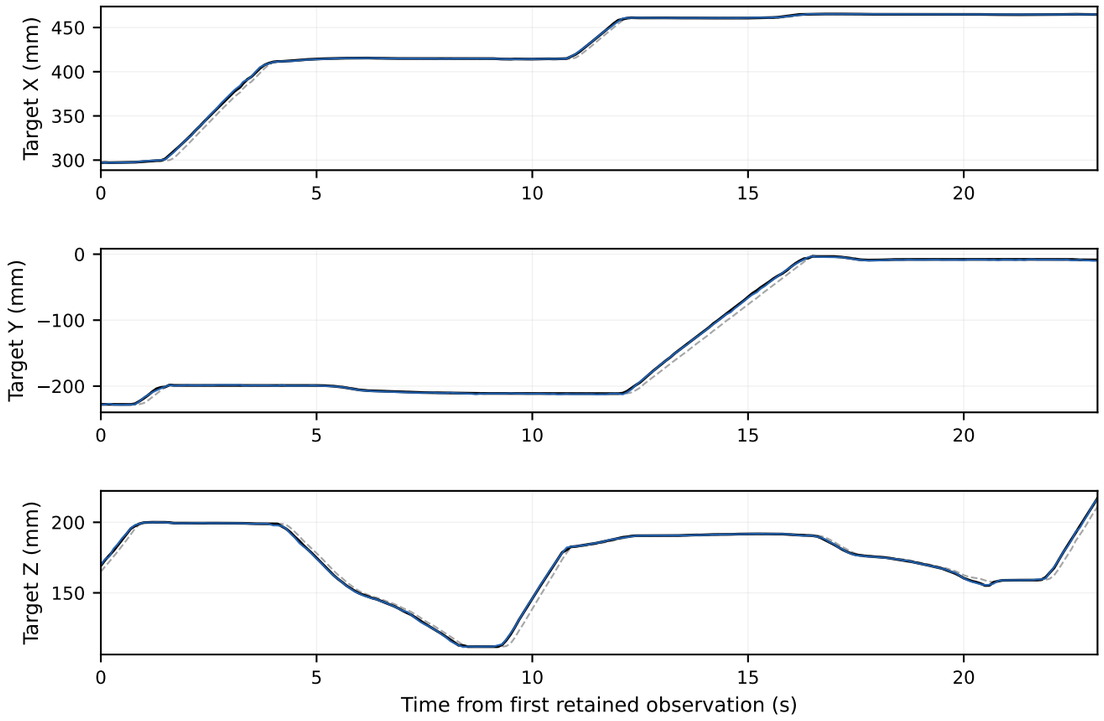
232 evaluated / 241 retained / 248 original frames | source starts at frame 7 | action rate 10 Hz

Mean XYZ target error: 0.813 mm | P95: 1.721 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 98.7%

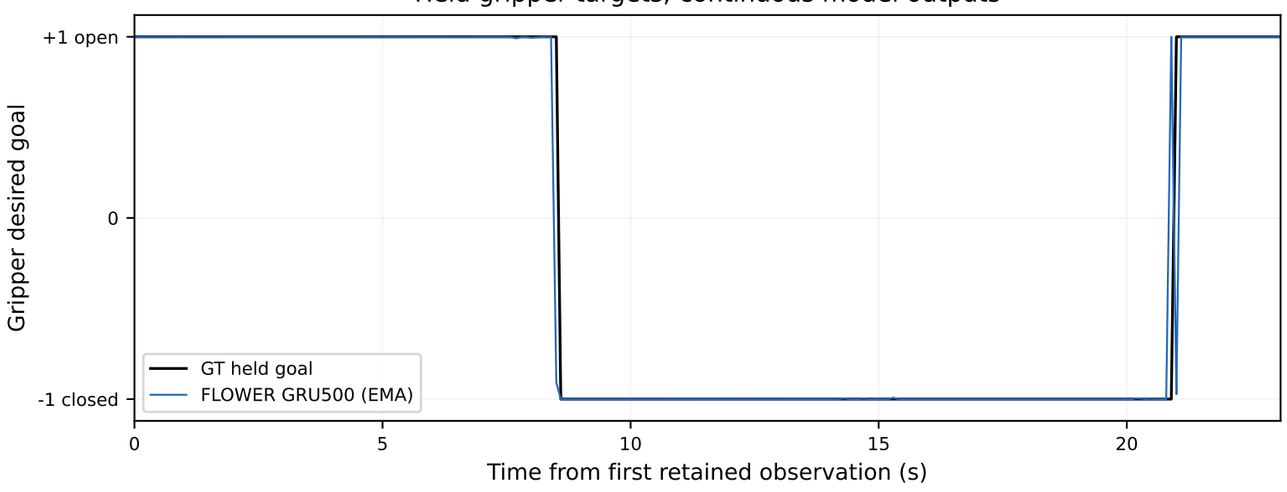
Recorded TCP and predicted command targets

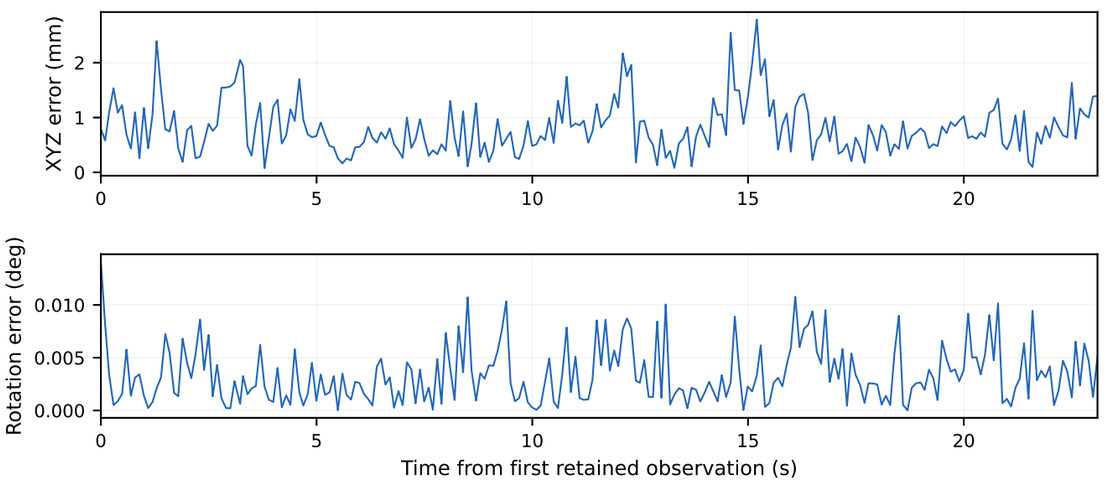


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

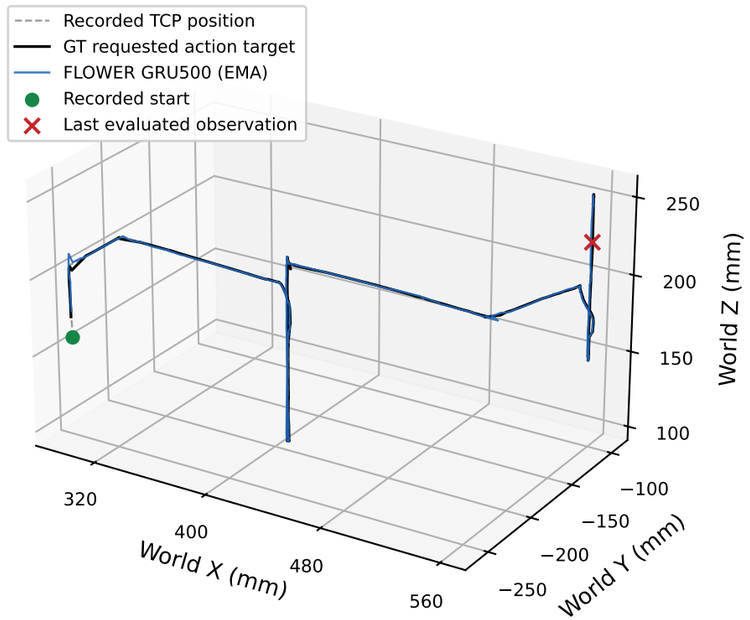
FLOWER GRU500 | lan p5 | episode 18/27 | RAW

1_demos_2026-10-23_17-14-13_767308.pkl

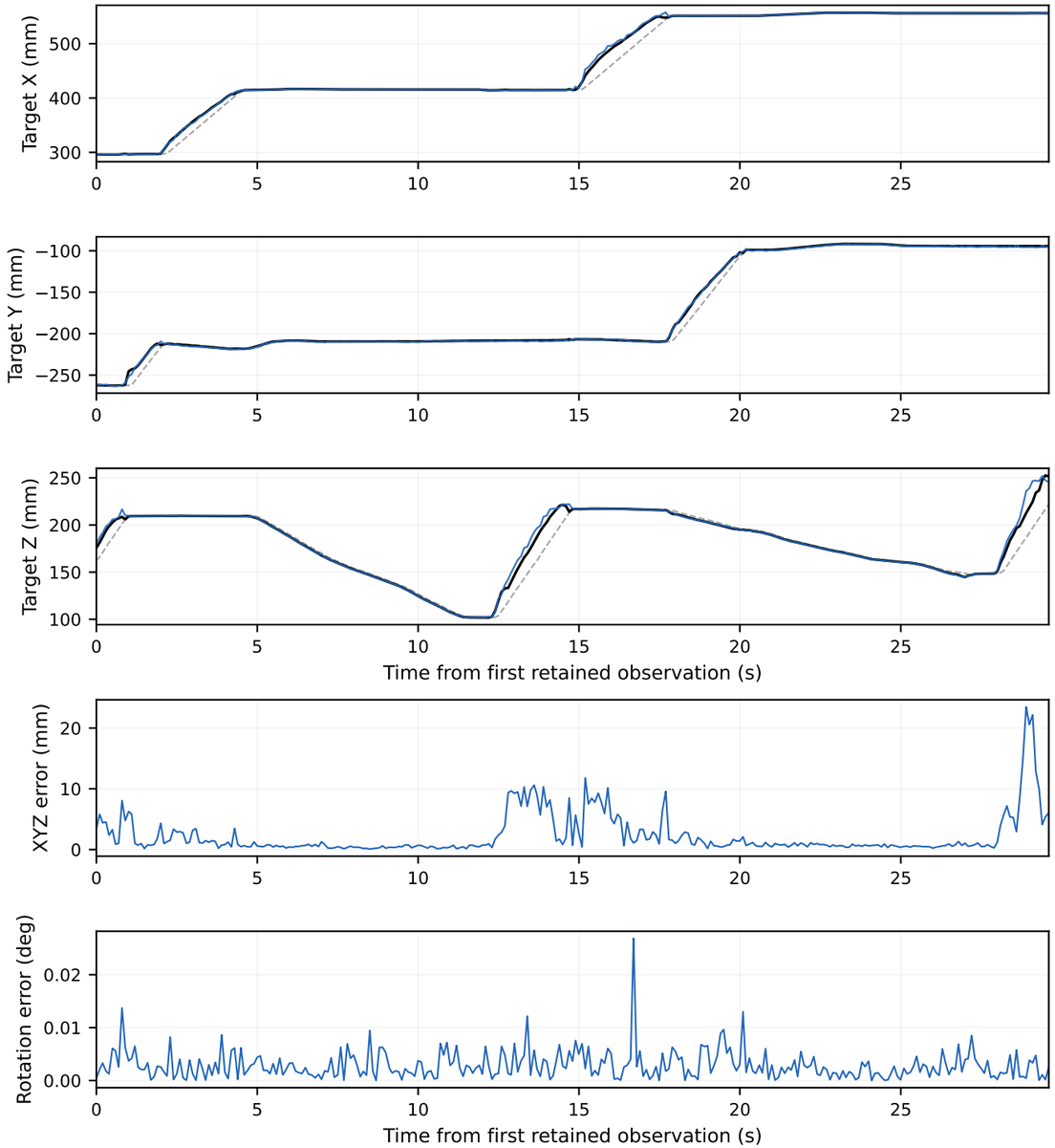
297 evaluated / 306 retained / 307 original frames | source starts at frame 1 | action rate 10 Hz

Mean XYZ target error: 2.279 mm | P95: 9.395 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 99.3%

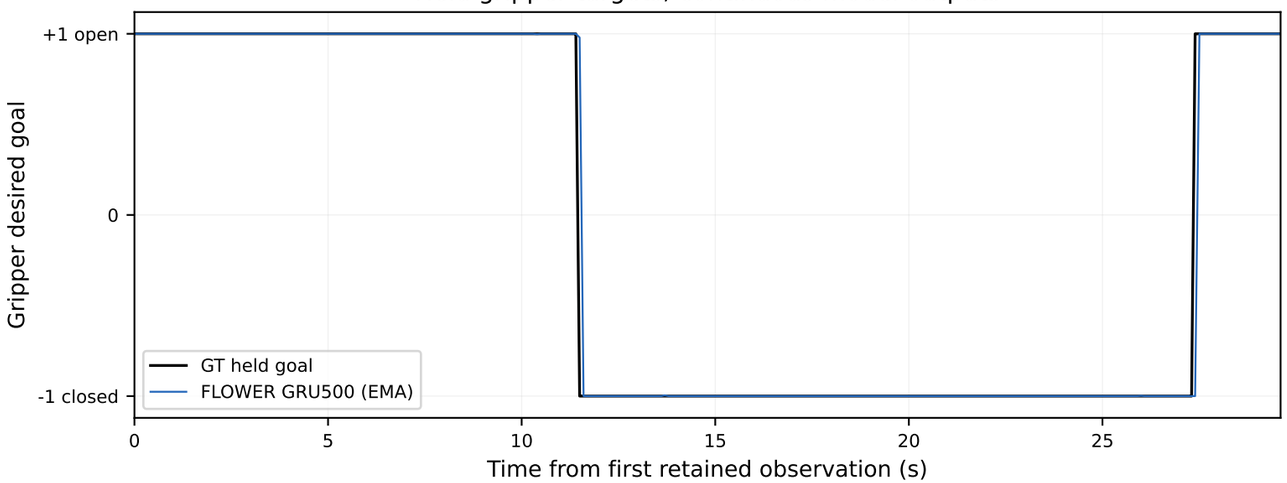
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



XYZ error (mm)

Rotation error (deg)

H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

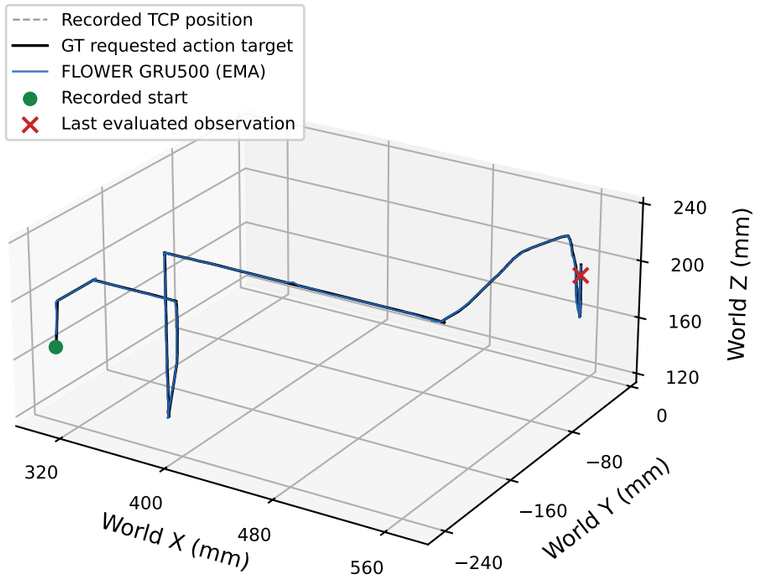
FLOWER GRU500 | outlet p1 | episode 19/27 | RAW

1_demos_2026-10-20_18-59-48_900578.pkl

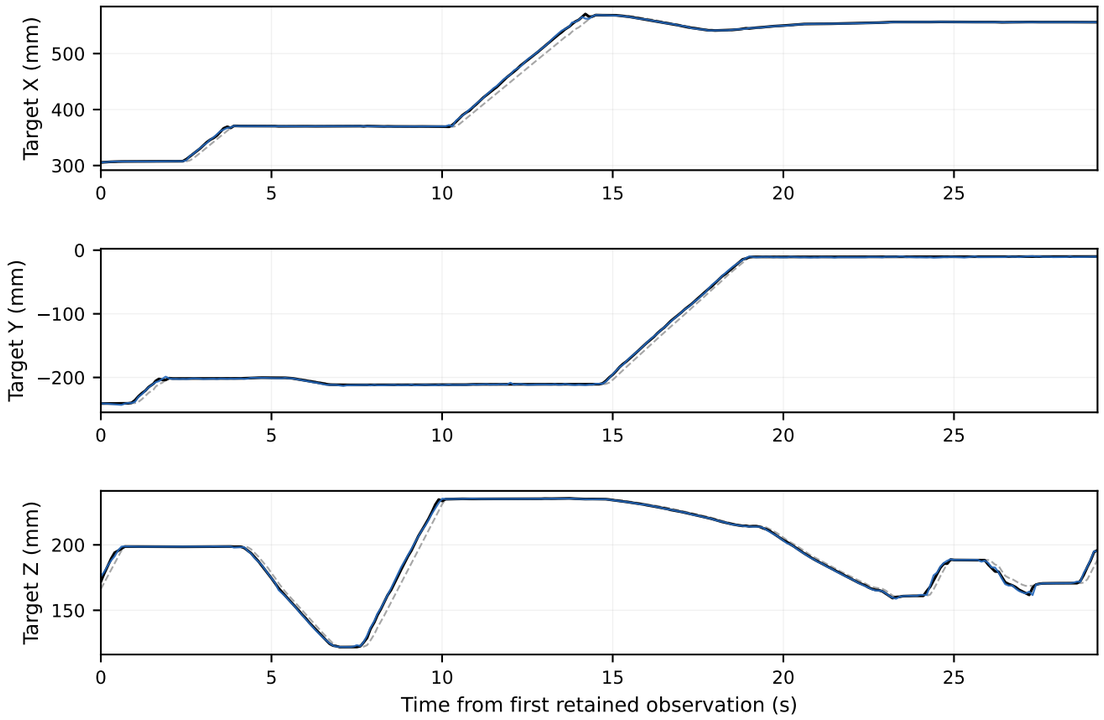
293 evaluated / 302 retained / 328 original frames | source starts at frame 26 | action rate 10 Hz

Mean XYZ target error: 1.059 mm | P95: 2.652 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 98.3%

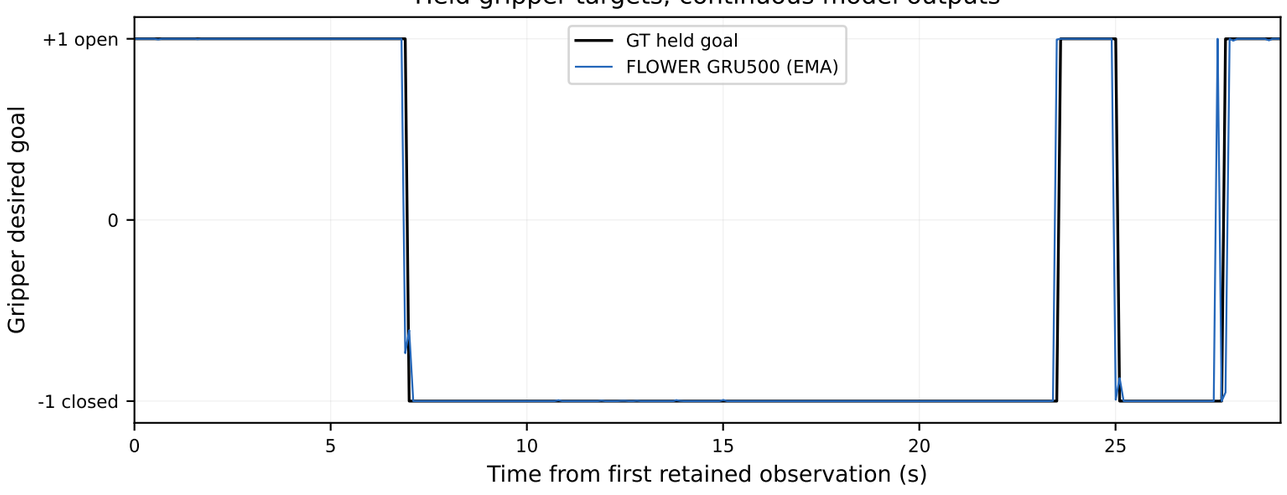
Recorded TCP and predicted command targets



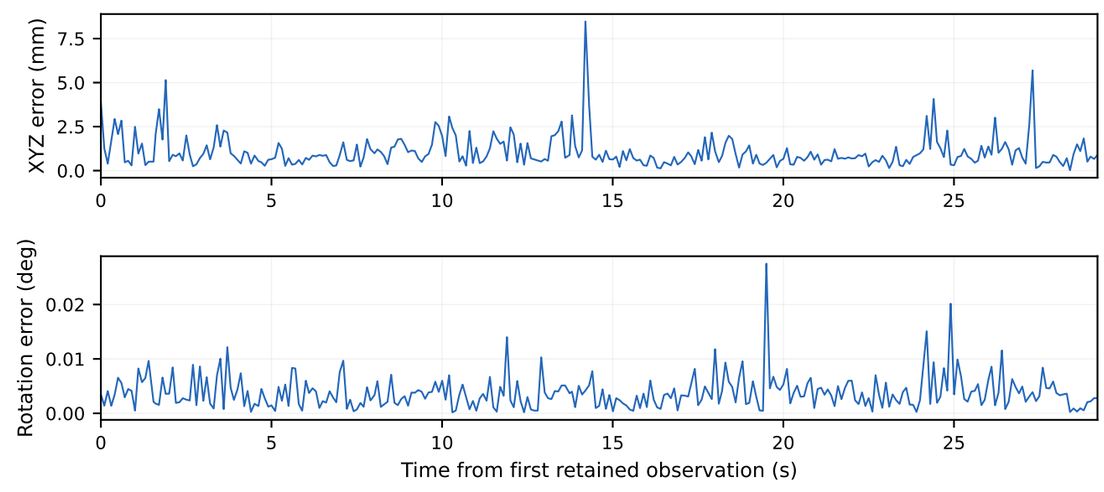
First command from each recorded training observation



Held gripper targets; continuous model outputs



Error



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

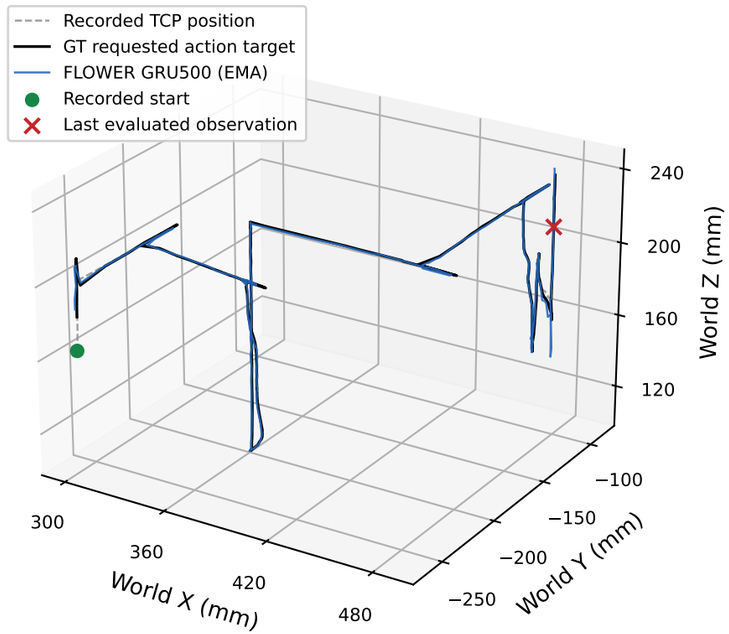
FLOWER GRU500 | outlet p3 | episode 20/27 | RAW

1_demos_2026-10-21_19-47-04_353401.pkl

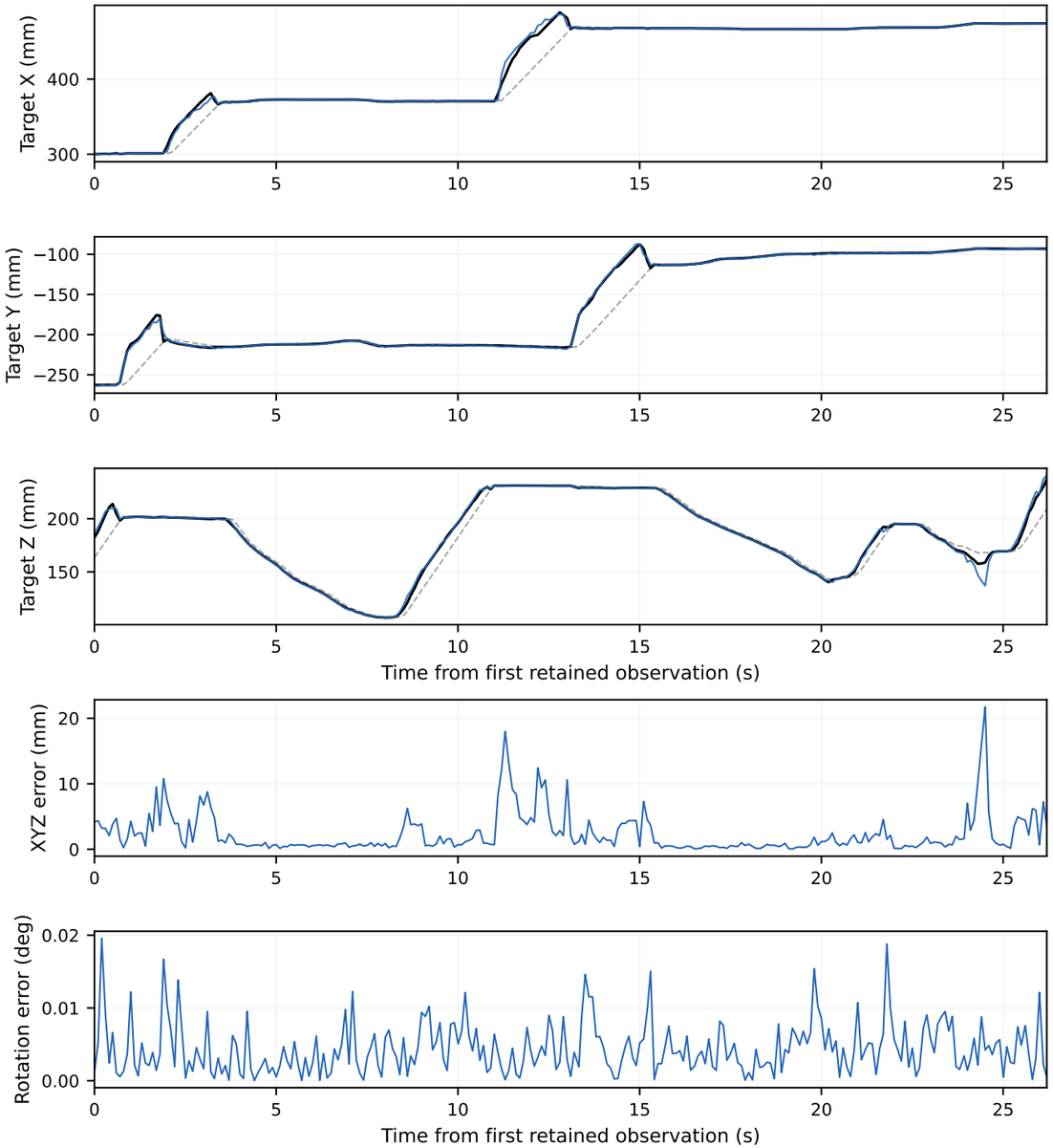
263 evaluated / 272 retained / 273 original frames | source starts at frame 1 | action rate 10 Hz

Mean XYZ target error: 2.372 mm | P95: 8.734 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 98.9%

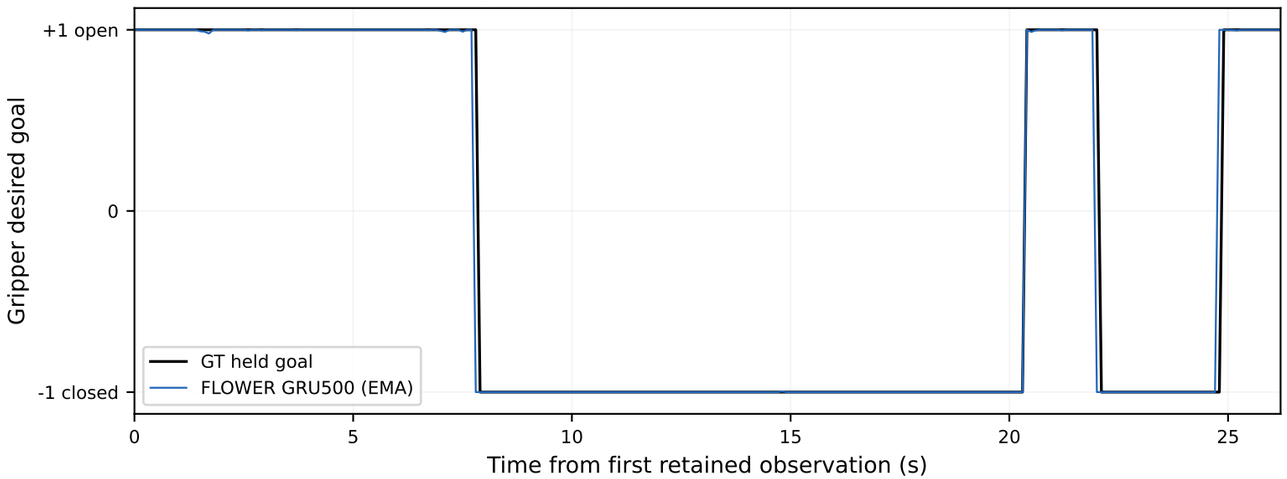
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

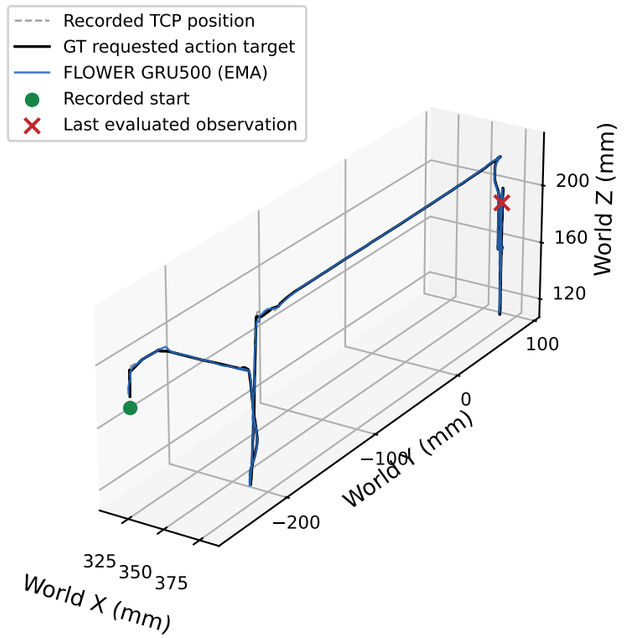
FLOWER GRU500 | outlet p5 | episode 21/27 | RAW

1_demos_2026-10-22_20-41-42_442405.pkl

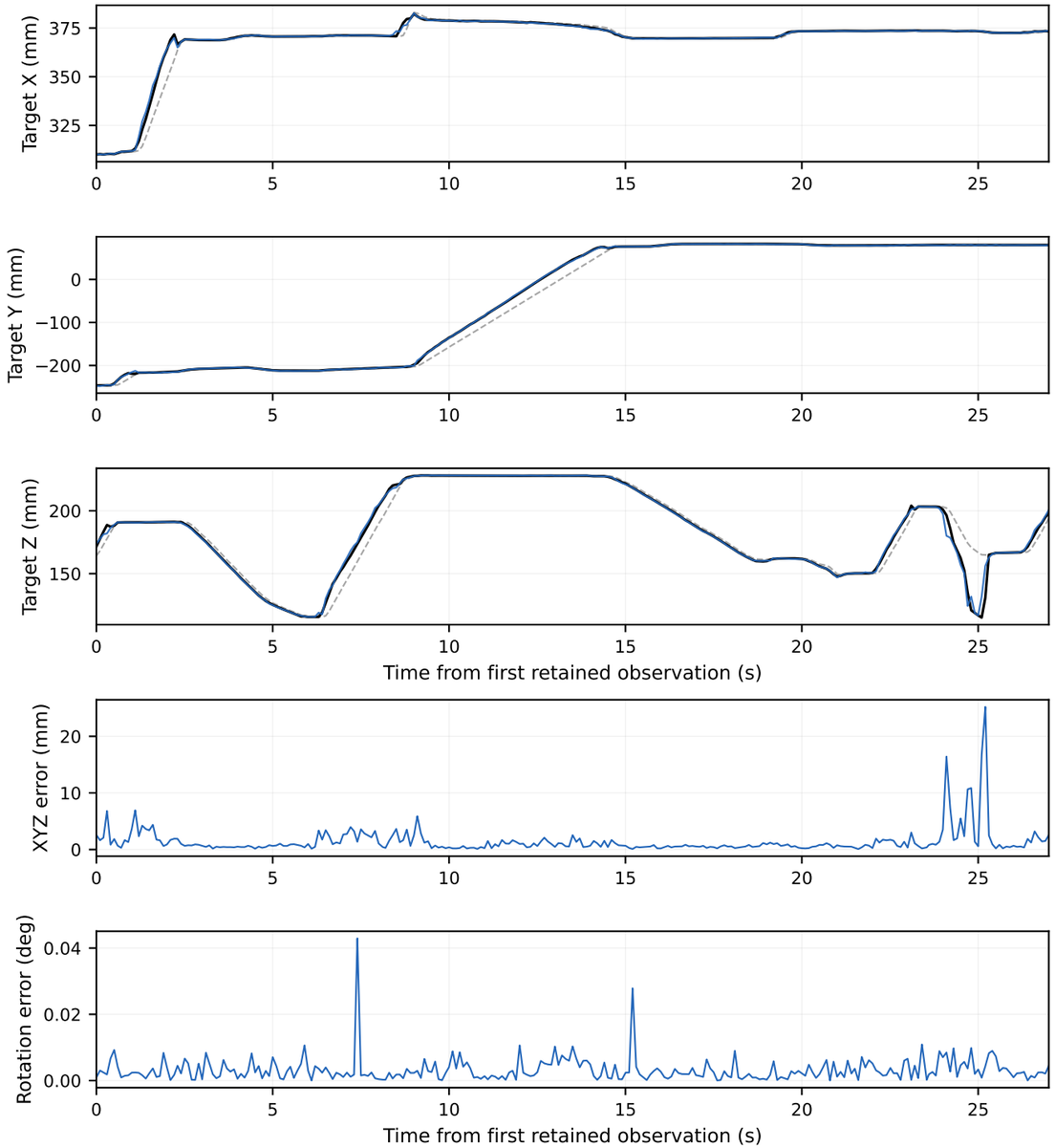
271 evaluated / 280 retained / 285 original frames | source starts at frame 5 | action rate 10 Hz

Mean XYZ target error: 1.429 mm | P95: 3.635 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 100.0%

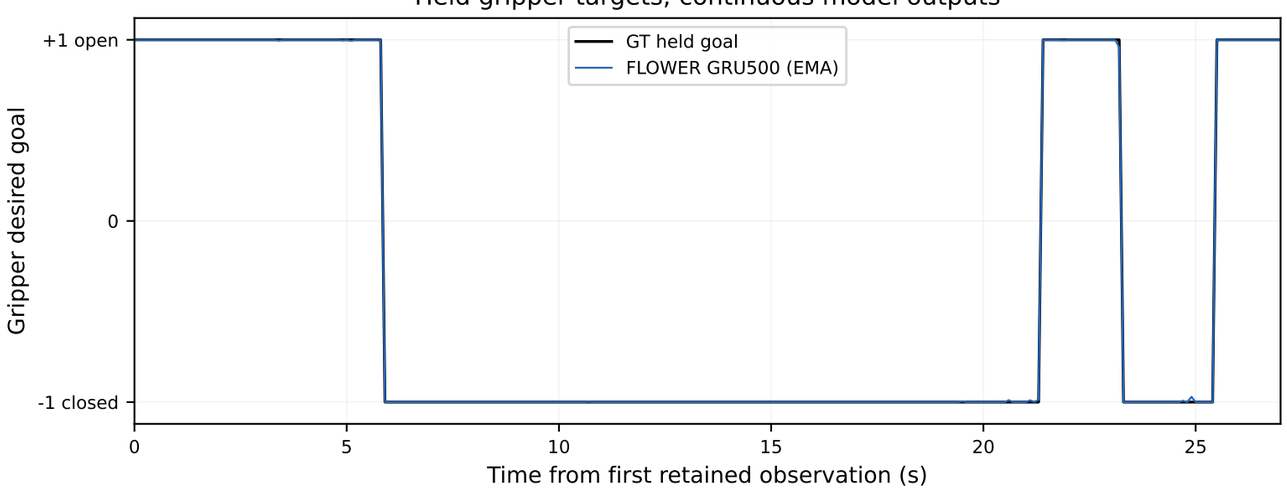
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



XYZ error (mm)

Rotation error (deg)

H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

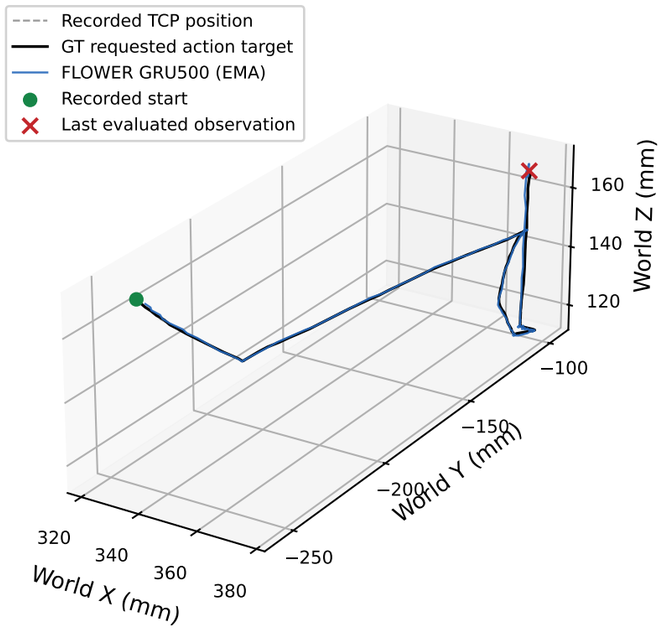
FLOWER GRU500 | rotary_switch p1 | episode 22/27 | RAW

1_demos_2026-10-19_17-16-02_422413.pkl

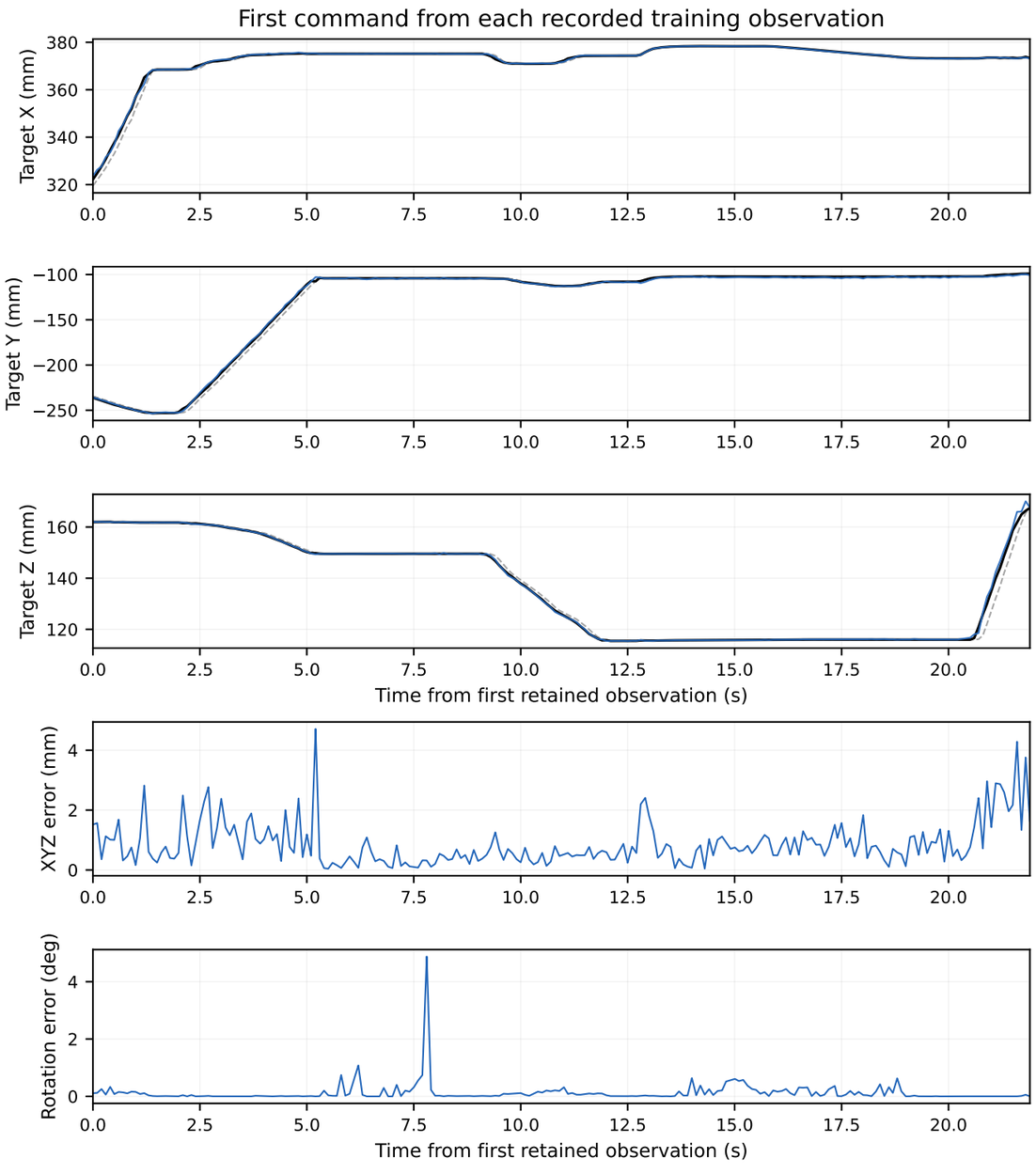
220 evaluated / 229 retained / 252 original frames | source starts at frame 23 | action rate 10 Hz

Mean XYZ target error: 0.859 mm | P95: 2.404 mm | mean rotation error: 0.125 deg | gripper sign accuracy: 97.3%

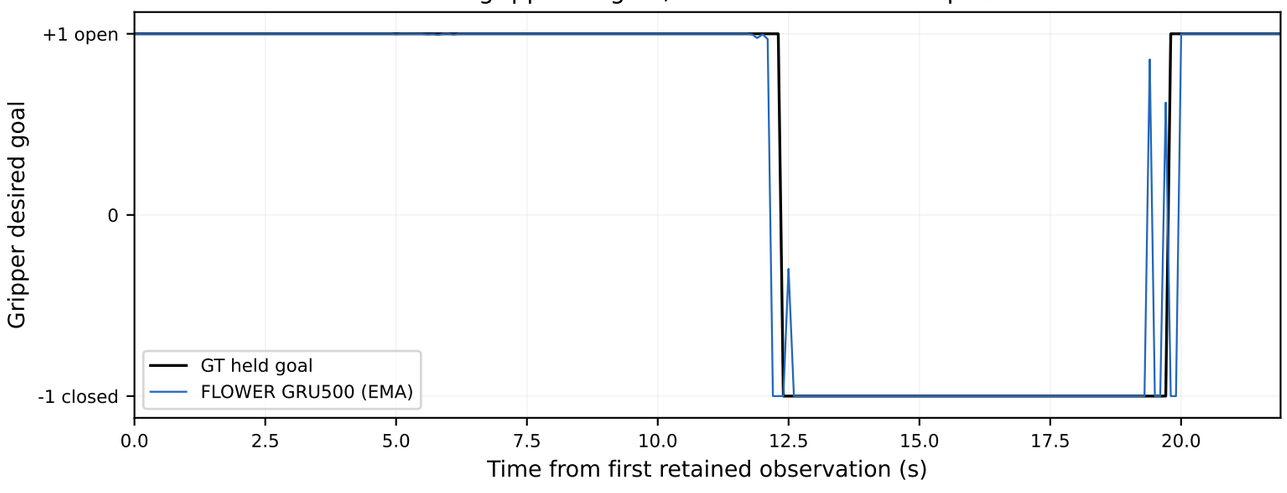
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

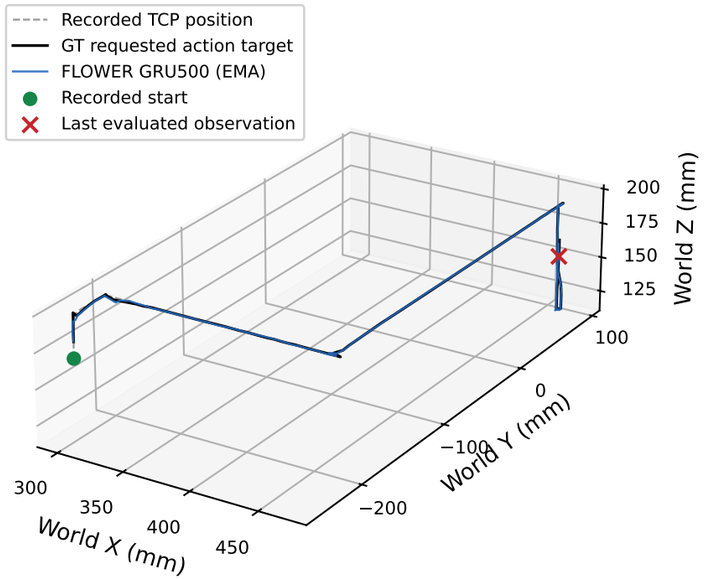
FLOWER GRU500 | rotary_switch p3 | episode 23/27 | RAW

1_demos_2026-10-22_16-58-35_836017.pkl

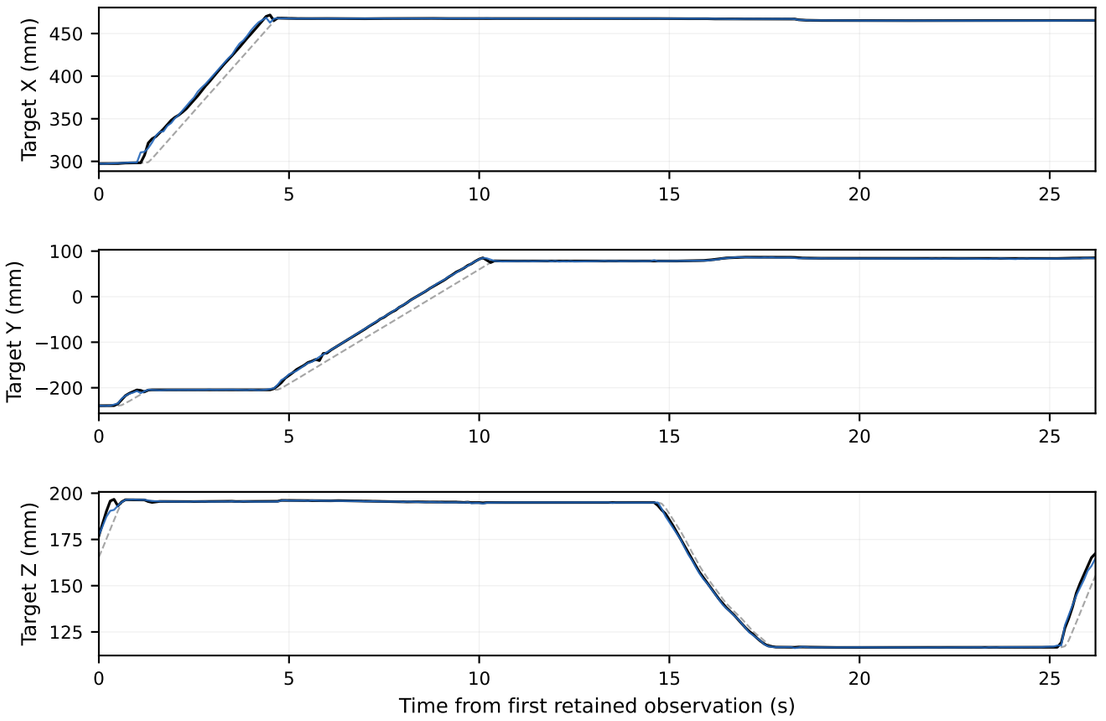
263 evaluated / 272 retained / 278 original frames | source starts at frame 6 | action rate 10 Hz

Mean XYZ target error: 1.225 mm | P95: 4.123 mm | mean rotation error: 0.128 deg | gripper sign accuracy: 100.0%

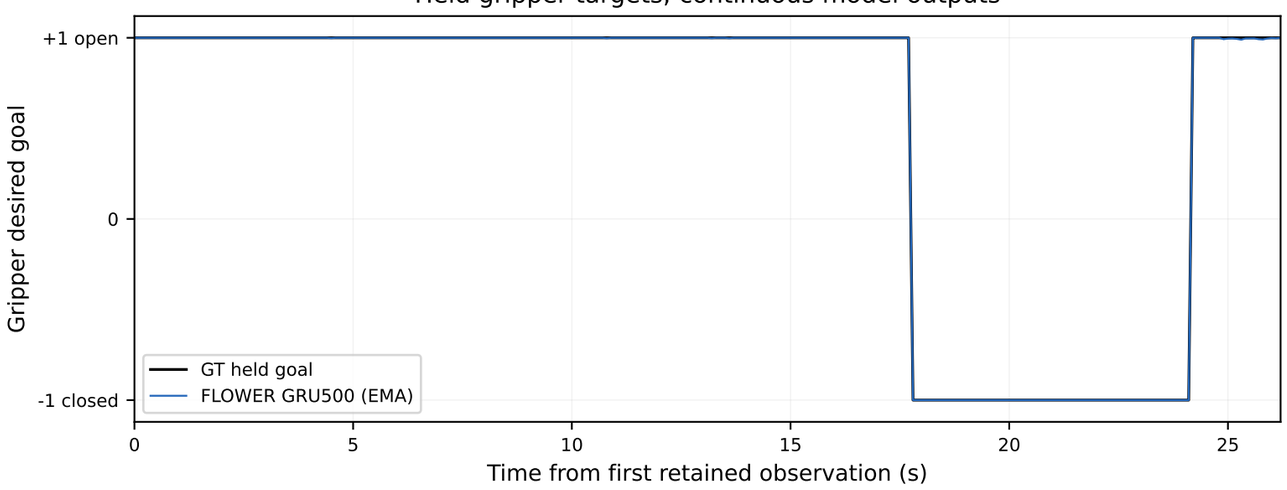
Recorded TCP and predicted command targets

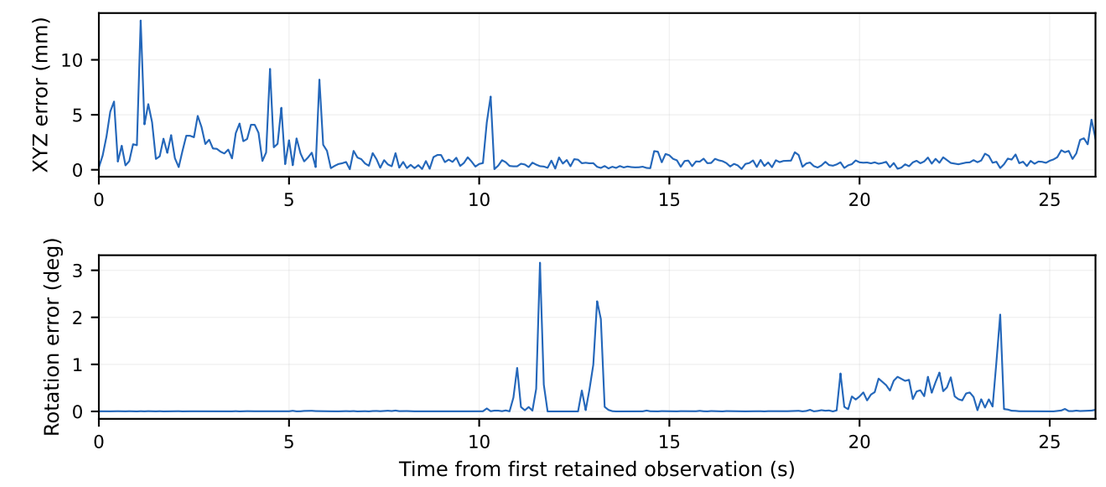


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

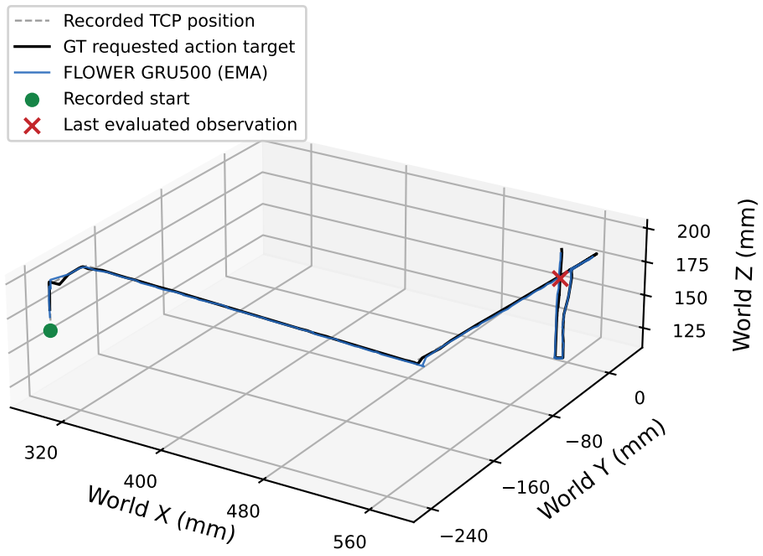
FLOWER GRU500 | rotary_switch p5 | episode 24/27 | RAW

1_demos_2026-10-23_17-26-43_291695.pkl

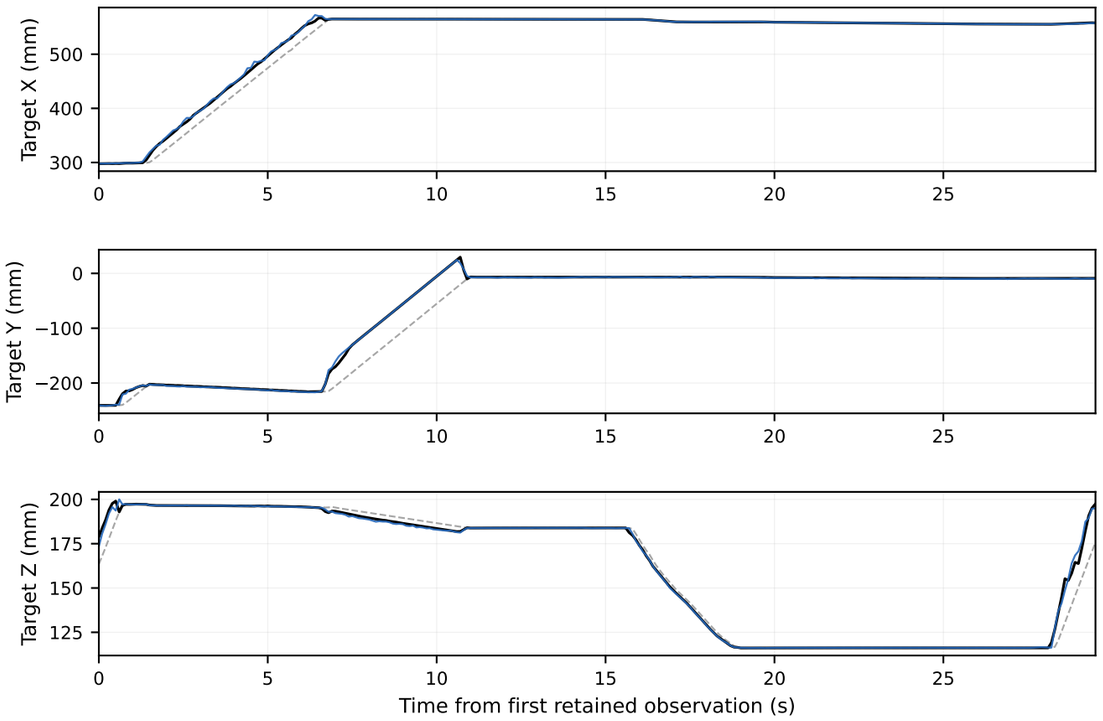
296 evaluated / 305 retained / 306 original frames | source starts at frame 1 | action rate 10 Hz

Mean XYZ target error: 1.561 mm | P95: 5.693 mm | mean rotation error: 0.075 deg | gripper sign accuracy: 98.6%

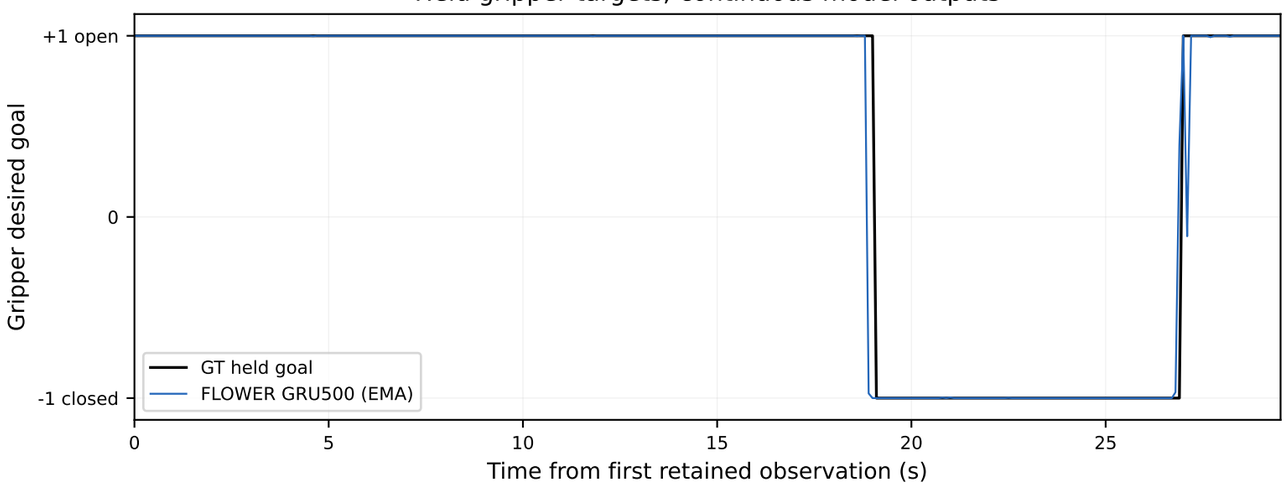
Recorded TCP and predicted command targets

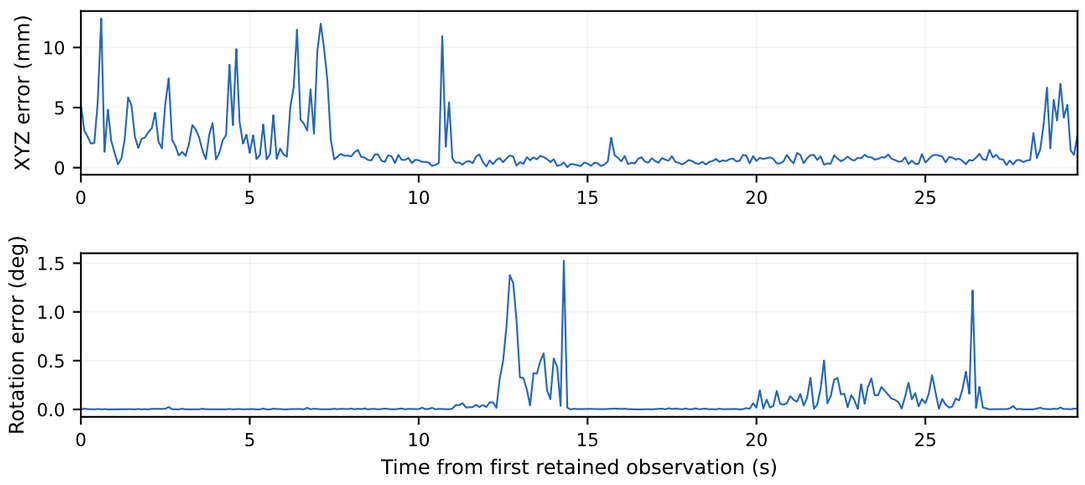


First command from each recorded training observation



Held gripper targets; continuous model outputs





H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

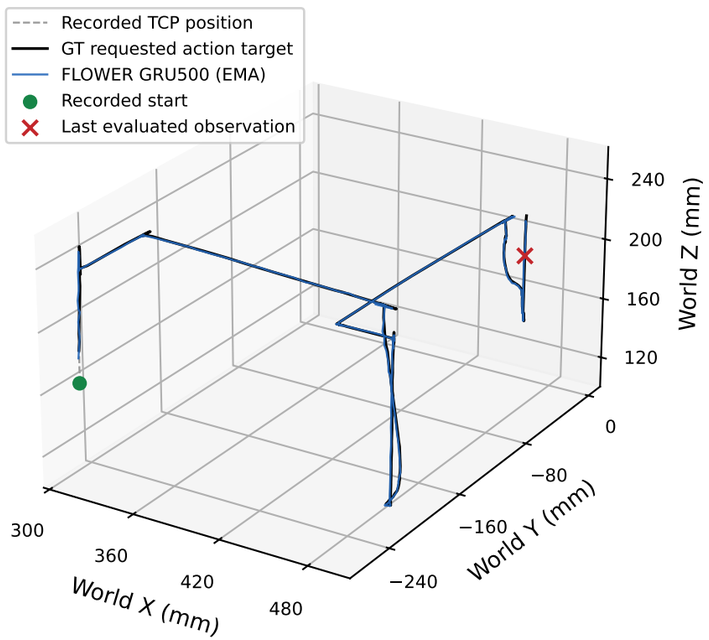
FLOWER GRU500 | usb p1 | episode 25/27 | RAW

1_demos_2026-10-20_18-10-52_902232.pkl

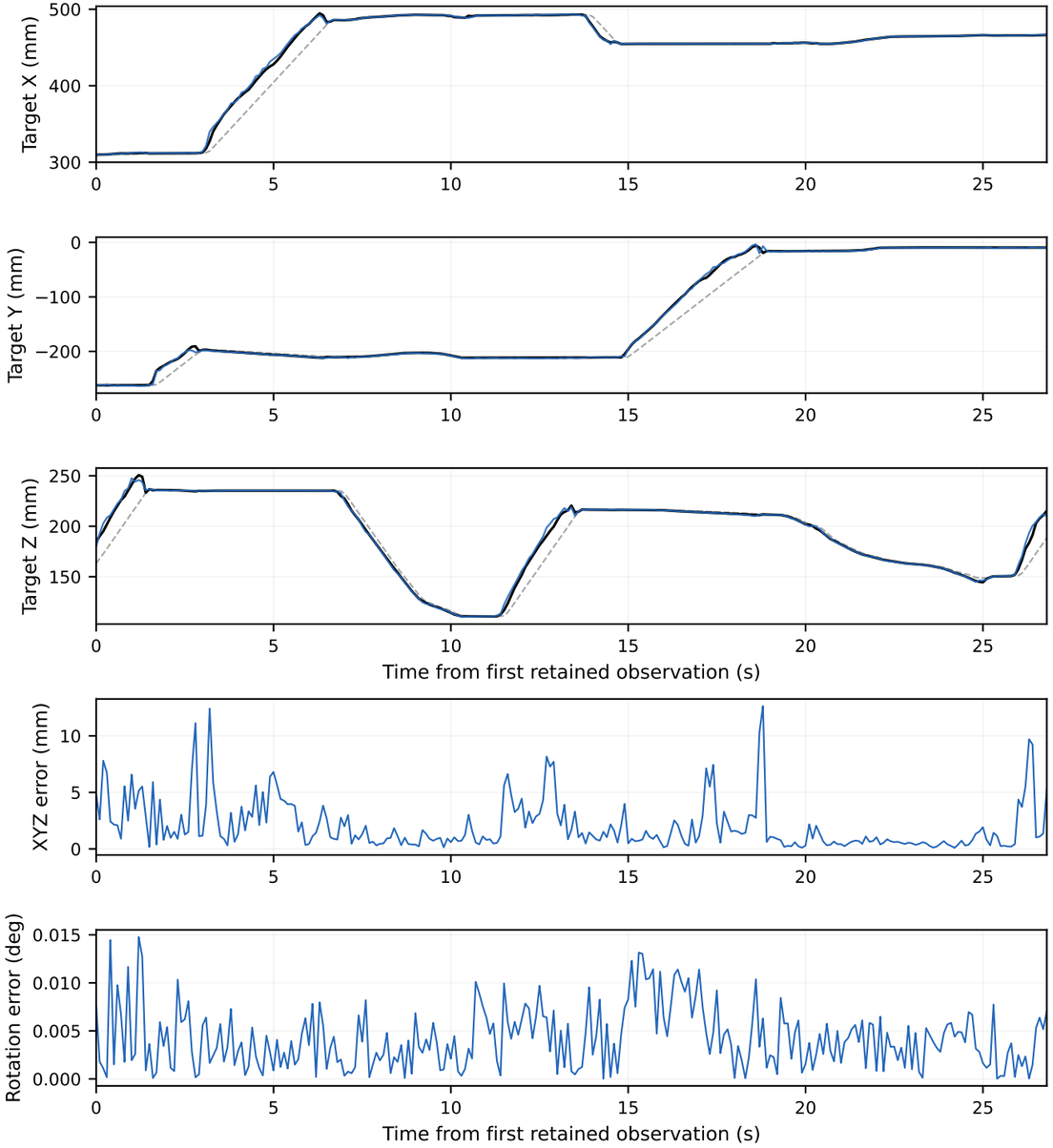
269 evaluated / 278 retained / 279 original frames | source starts at frame 1 | action rate 10 Hz

Mean XYZ target error: 2.048 mm | P95: 6.779 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 98.9%

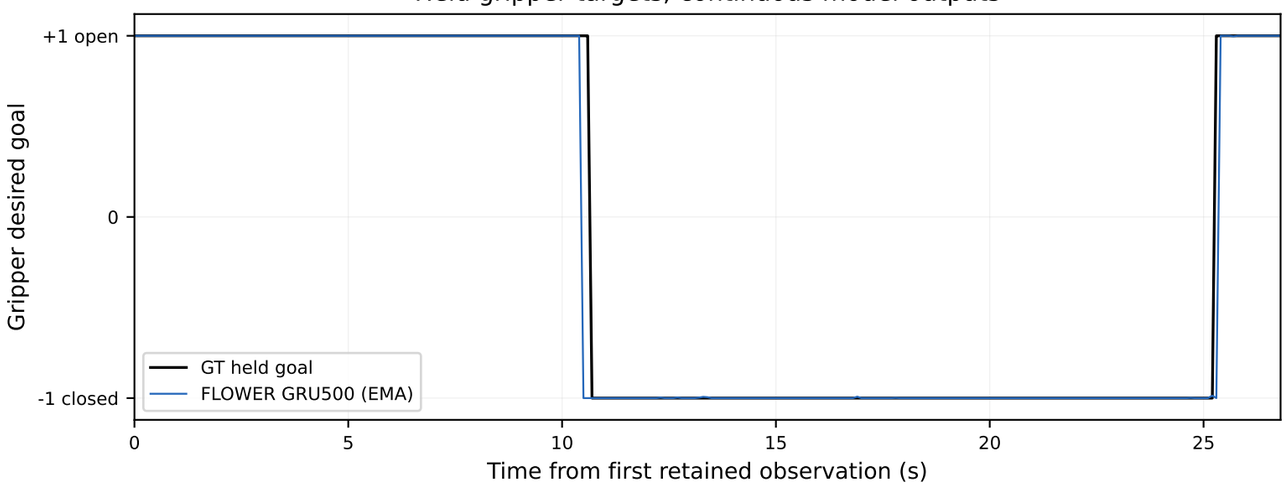
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



XYZ error (mm)

Rotation error (deg)

H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

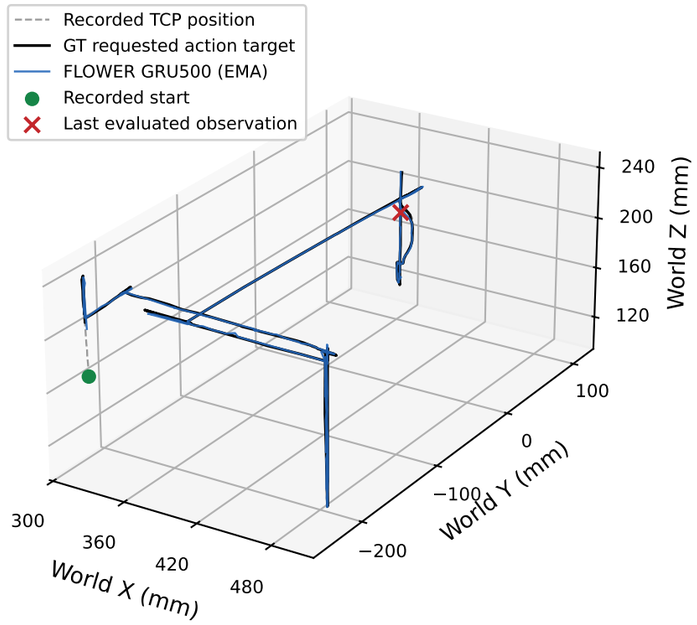
FLOWER GRU500 | usb p3 | episode 26/27 | RAW

1_demos_2026-10-21_18-00-45_456589.pkl

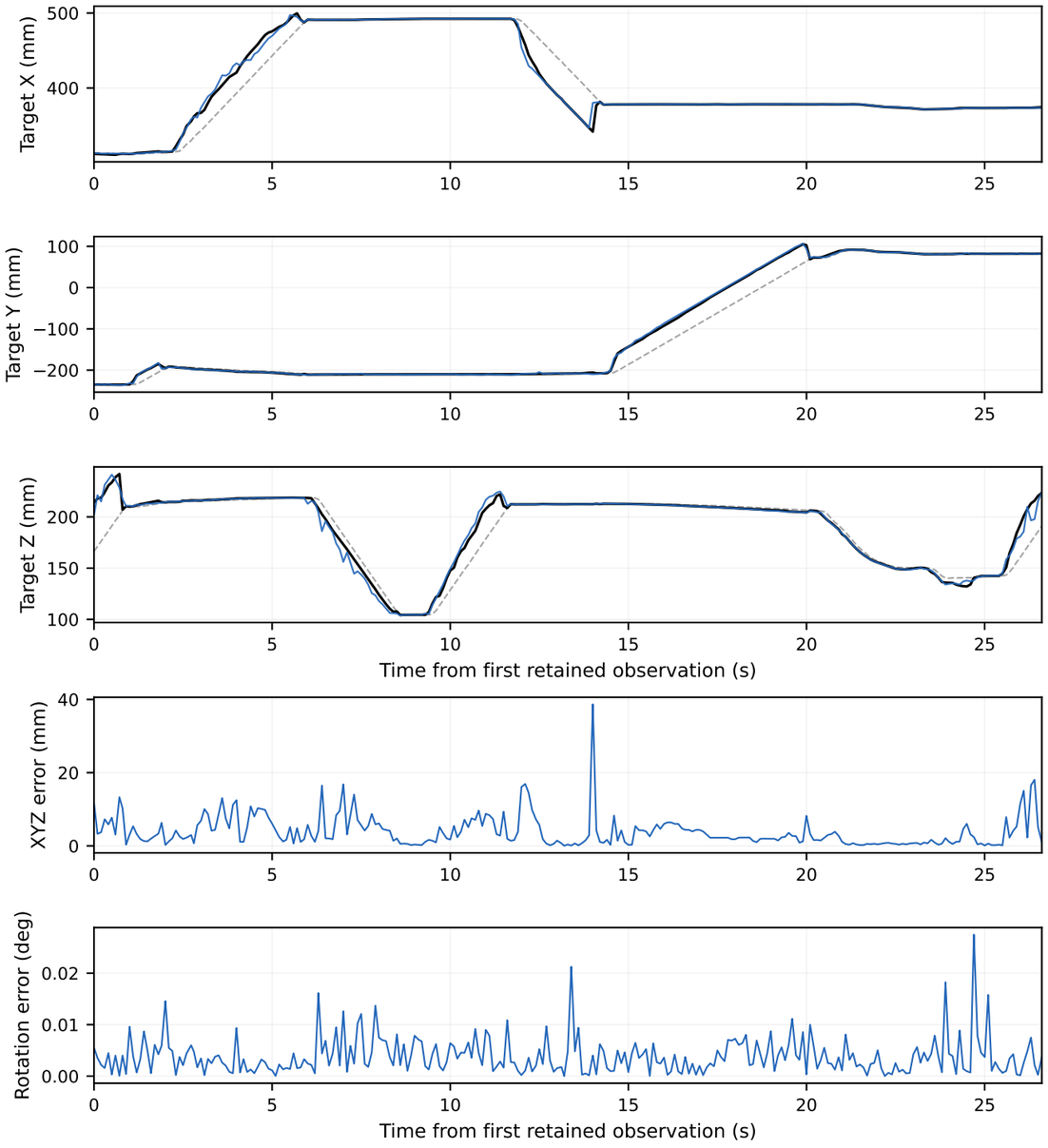
267 evaluated / 276 retained / 278 original frames | source starts at frame 2 | action rate 10 Hz

Mean XYZ target error: 3.931 mm | P95: 11.665 mm | mean rotation error: 0.004 deg | gripper sign accuracy: 99.6%

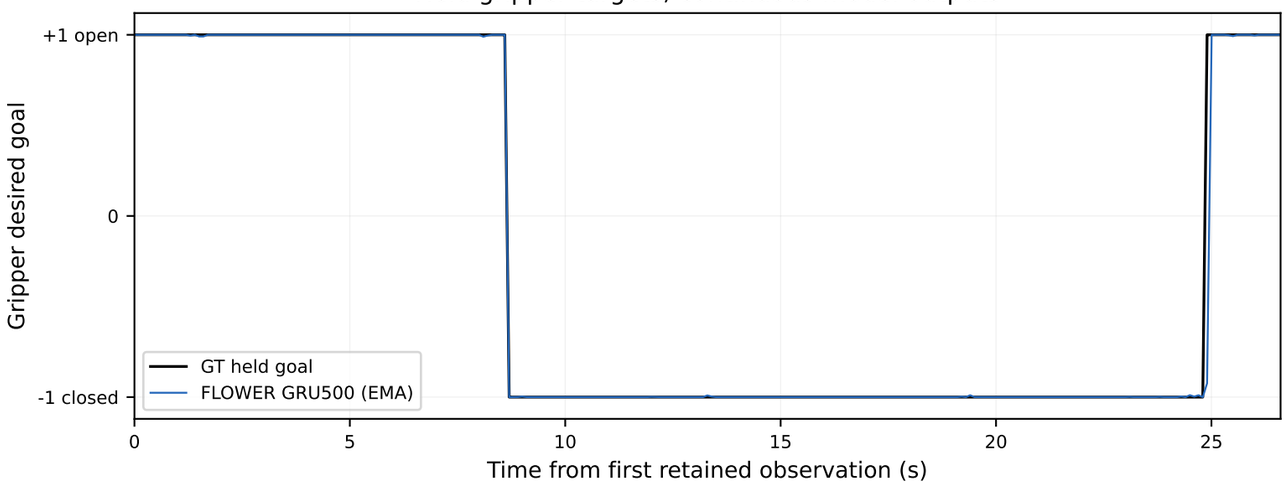
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



XYZ error (mm)

Rotation error (deg)

H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation

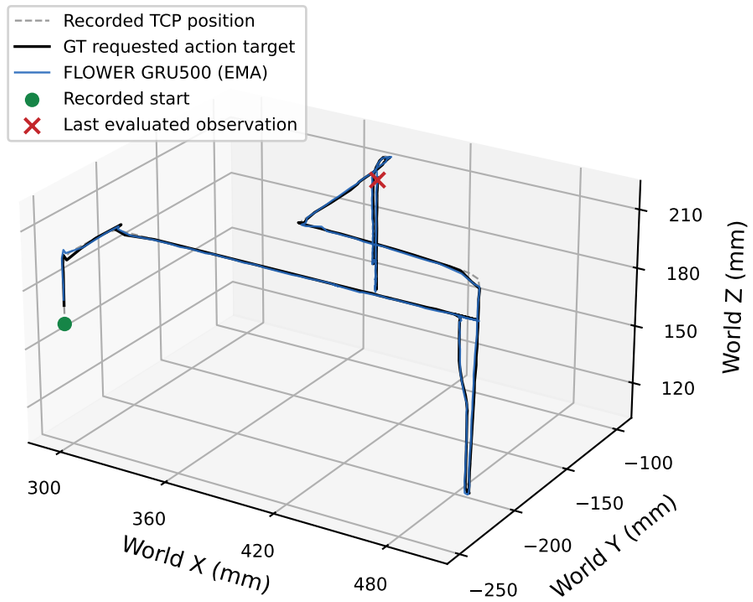
FLOWER GRU500 | usb p5 | episode 27/27 | RAW

1_demos_2026-10-22_20-27-01_776940.pkl

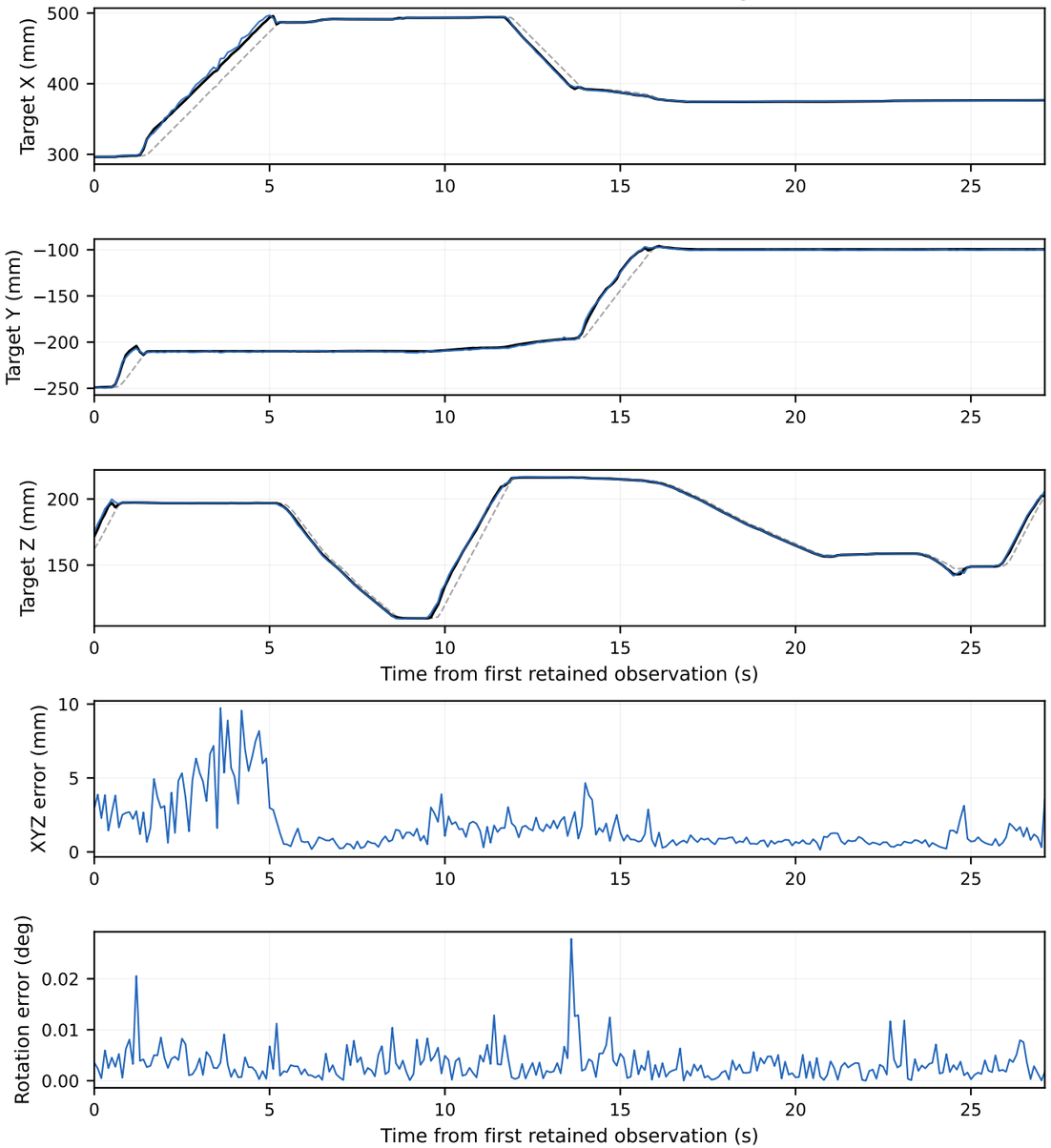
272 evaluated / 281 retained / 287 original frames | source starts at frame 6 | action rate 10 Hz

Mean XYZ target error: 1.683 mm | P95: 5.405 mm | mean rotation error: 0.003 deg | gripper sign accuracy: 99.3%

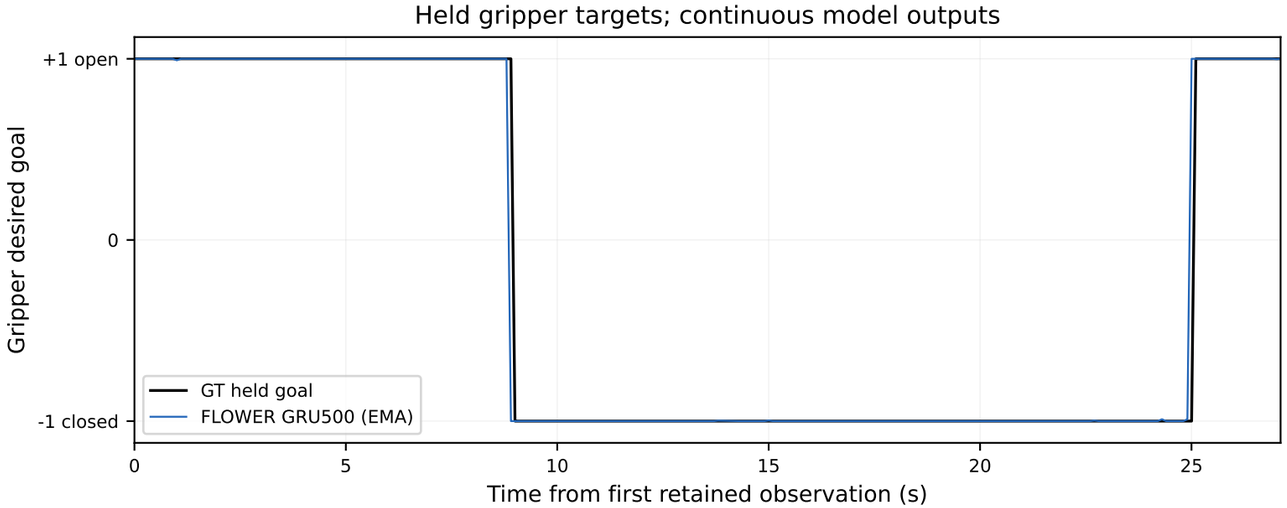
Recorded TCP and predicted command targets



First command from each recorded training observation



Held gripper targets; continuous model outputs



H=10; first action at every complete training-window start. Targets = recorded TCP + decoded base-frame delta; rotations use left multiplication.

Recorded RGB, 13-D state, task instruction and 500 Hz FT history. Recorded-observation replay; no cumulative integration, robot rollout or output smoothing.

CPU FP32 | 4 Euler sampling steps | EMA epoch 20 @ 20,820 steps | last H-1 observations excluded | training data, not a held-out evaluation